

Journal of Mathematical Economics

Additive Measurement on Path-Generated Domains: Directed Recovery in Rational Addiction --Manuscript Draft--

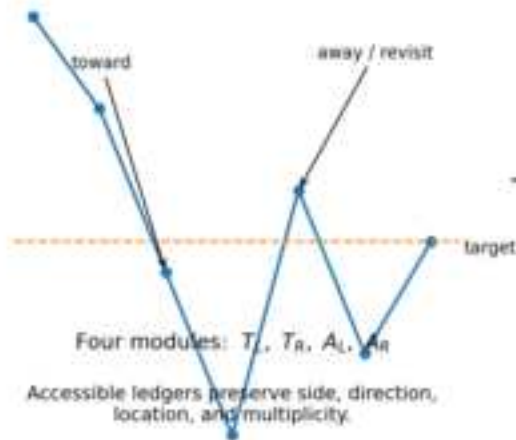
Manuscript Number:	
Article Type:	Research Paper
Keywords:	rational addiction; directed recovery; target-relative path; finite cancellation; Stieltjes representation; target projection
Corresponding Author:	Yushang Cheng Southwest Jiaotong University Hope College CHINA
First Author:	Yushang Cheng
Order of Authors:	Yushang Cheng
Abstract:	This paper studies an additive measurement problem on a path-generated domain that is not closed under addition. The primitive object is a single continuous target-relative experience, represented by a four-module ledger of toward-target and away-from-target passages on the two sides of a target. Rational addiction supplies the leading economic application: conditional on a predetermined, nonbinding target schedule, the ledger evaluates recovery motion in Becker–Murphy type stock dynamics. On the complete domain generated by continuous paths that are monotone on a finite partition, even the formal-increment version of location-free common-increment cancellation (LC-A) does not imply finite cancellation. LC-A is a tempting bridge because it permits every common formal shift for which both translated ledgers remain accessible. The counterexample shows why that bridge fails: path accessibility can protect an endpoint-flow tie breaker from all admissible common shifts while still permitting a balanced finite list of strict comparisons. The positive theorem is a separate sufficiency result, not a minimal behavioral characterization. Under same-ledger indifference, a weak order, finite cancellation, difference completeness, list Archimedean regularity, strict directional monotonicity, and path-generated monotone continuity, the order admits an exact four-module Stieltjes representation. The measures may contain atoms, including target-arrival atoms under a directed-arrival convention. In the absolutely continuous subclass, the representation enters BM dynamics through a recovery wedge that may strengthen, weaken, or reverse primitive adjacent complementarity. Two explicitly one-step illustrations show how strong target pull selects the feasible state closest to the target, producing projection plateaus and, under conditional iteration of the same map, one-sided finite-time capture of sustainable positive targets. The path-domain characterization, the separation of LC-A from finite cancellation, and the sufficient Stieltjes representation route are the paper's primary theoretical contributions.

Directed Recovery on a Path-Generated Domain

A cancellation boundary, a sufficient Stieltjes representation, and a leading addition application

1. Primary object

One continuous target-relative path



The domain is generated by one connected path and is not closed under formal addition.

2. Boundary and representation

$LC-\mathcal{K} \Rightarrow$ finite cancellation

A protected endpoint-flow state survives every admissible common translation.

$FC + DC + LA + PGMC^+$

$\Downarrow$

$$U(Y) = \int h^{T_L} d\lambda_{T_L} + \int h^{T_R} d\lambda_{T_R} - \int h^{A_L} d\lambda_{A_L} - \int h^{A_R} d\lambda_{A_R}$$

Canonical ordered group + four-module Lebesgue-Stieltjes representation

A sufficient route under stated order and continuity conditions,

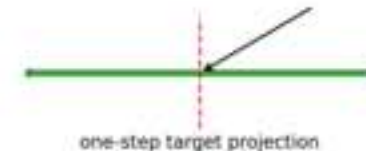
3. Leading economic application

Predetermined, nonbinding target schedule

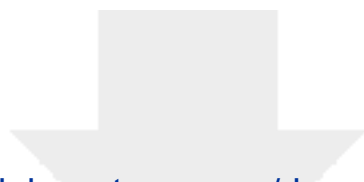
$$A_{t+1} = (1 - \delta)A_t + c_t$$

$$W_{CA} = u_{CA} + \eta R_d$$

The recovery wedge may strengthen, weaken, or reverse primitive adjacent complementarity.



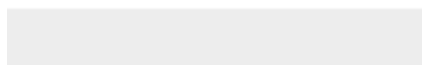
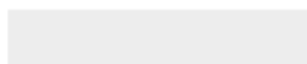
Conditional one-step illustrations: recovery wedge and target projection.



[Click here to access/download](#)

Supplementary material

[Declaration_of_Competing_Interest.docx](#)



13 July 2026

Dear Editors,

Please consider the manuscript entitled *Additive Measurement on Path-Generated Domains: Directed Recovery in Rational Addiction* for publication in the *Journal of Mathematical Economics*. The paper's primary object is an additive measurement problem on a path-generated domain that is not closed under addition. Rational addiction supplies the leading economic application rather than the source of the representation theorem.

A single continuous target-relative path generates a four-module ledger of toward-target and away-from-target passages on the two sides of a target. Because accessible ledgers must arise from one connected path, standard semigroup completion and unrestricted cancellation arguments are unavailable. The paper shows that even the formal-increment version of location-free common-increment cancellation, together with strict directional monotonicity, does not imply finite cancellation. The counterexample is defined on the full continuous path-generated domain and provides a balanced finite list of strict comparisons whose aggregate ledgers coincide.

The positive theorem is logically separate from the counterexample. Starting from finite cancellation as an independent finite-list consistency condition, it constructs a canonical ordered group; under difference completeness, list Archimedean regularity, and path-generated monotone continuity, it obtains an exact four-module Lebesgue-Stieltjes representation. This is presented as a transparent sufficient route rather than a minimal behavioral characterization. The paper also characterizes finite-step ledger reachability, states the exact uniqueness class, and isolates the restrictions created by target atoms when arbitrarily short return loops are feasible.

The Becker-Murphy stock dynamics provide the principal economic interpretation. Conditional on a predetermined, nonbinding target schedule, directed recovery adds a process component that evaluates movement toward and away from the target. Two explicitly one-step illustrations show how the resulting recovery wedge can modify primitive adjacent complementarity and how sufficiently strong target pull can select the feasible next state closest to the target. These sections are not presented as a Bellman-optimal rational-addiction policy theorem; endogenous target formation and a full recursive model are left for separate analysis.

I believe the manuscript fits the *Journal of Mathematical Economics* because its central contribution combines a nonstandard economically interpretable choice domain with finite cancellation, ordered-group methods, Stieltjes representation, and an explicit boundary counterexample. The manuscript is original, is not under consideration elsewhere, and uses no empirical data. The graphical abstract, highlights, and declaration of competing interest are submitted separately.

Thank you for your consideration.

Sincerely,

Yushang Cheng
Business School, Southwest Jiaotong University Hope College
Chengdu, Sichuan, China
hegebittel@gmail.com

- A single continuous path generates a non-additively-closed ledger domain.
- Location-free common increments do not imply finite cancellation.
- Finite cancellation yields a canonical ordered-group representation.
- Regularity gives a four-module Stieltjes representation.
- One-step illustrations link target motion to recovery wedges and projection.

Additive Measurement on Path-Generated Domains: Directed Recovery in Rational Addiction

Yushang Cheng^{a,*}

^a*Business School, Southwest Jiaotong University Hope College, Chengdu, China*

ARTICLE INFO

Keywords:

rational addiction
directed recovery
target-relative path
finite cancellation
Stieltjes representation
target projection

ABSTRACT

This paper studies an additive measurement problem on a path-generated domain that is not closed under addition. The primitive object is a single continuous target-relative experience, represented by a four-module ledger of toward-target and away-from-target passages on the two sides of a target. Rational addiction supplies the leading economic application: conditional on a predetermined, nonbinding target schedule, the ledger evaluates recovery motion in Becker–Murphy type stock dynamics.

On the complete domain generated by continuous paths that are monotone on a finite partition, even the formal-increment version of location-free common-increment cancellation (LC-A) does not imply finite cancellation. LC-A is a tempting bridge because it permits every common formal shift for which both translated ledgers remain accessible. The counterexample shows why that bridge fails: path accessibility can protect an endpoint-flow tie breaker from all admissible common shifts while still permitting a balanced finite list of strict comparisons. The positive theorem is a separate sufficiency result, not a minimal behavioral characterization. Under same-ledger indifference, a weak order, finite cancellation, difference completeness, list Archimedean regularity, strict directional monotonicity, and path-generated monotone continuity, the order admits an exact four-module Stieltjes representation. The measures may contain atoms, including target-arrival atoms under a directed-arrival convention.

In the absolutely continuous subclass, the representation enters BM dynamics through a recovery wedge that may strengthen, weaken, or reverse primitive adjacent complementarity. Two explicitly one-step illustrations show how strong target pull selects the feasible state closest to the target, producing projection plateaus and, under conditional iteration of the same map, one-sided finite-time capture of sustainable positive targets. The path-domain characterization, the separation of LC-A from finite cancellation, and the sufficient Stieltjes representation route are the paper's primary theoretical contributions.

JEL classification: C61; D11; D90; I12.

1. Introduction

This paper's primary object is an additive measurement problem on a path-generated domain that is not closed under addition. Each primitive alternative is a single continuous experience relative to a target. Its directional ledger records where the path moves toward or away from the target, on which side, and with what multiplicity. Rational addiction supplies the leading economic application, rather than the source of the representation theorem.

The strength of rational-addiction theory is that it places behavior that appears uncontrolled inside a clear dynamic-choice structure. In the benchmark model of [Becker and Murphy \(1988\)](#), current consumption forms a stock that changes future marginal utility, and adjacent complementarity can amplify present choices. [Becker et al. \(1994\)](#) subsequently brought this structure to empirical demand analysis. What the benchmark does not isolate is a preference over the process of recovery itself. A low stock or low consumption may accompany recovery, but neither identifies whether the state is moving toward an intended position, moving away from it, overshooting it, or repeatedly crossing it.

The leading application therefore introduces a target state A^* and treats recovery as target-relative directed motion. Given the actual state A , the coordinate $x = \Psi(A, A^*)$ places the state relative to the target. An action induces a path from the current coordinate to the next one. The path may approach the target, cross it, leave it, reverse direction, and revisit the same interval. Recovery evaluation records this directed motion rather than only terminal distance. Exit is

*Corresponding author

 heggebittel@gmail.com (Y. Cheng)

ORCID(s): 0009-0001-3218-6423 (Y. Cheng)

the special case $A^* = 0$; positive targets describe reduction, maintenance, or treatment plateaus. The recovery ranking is not a complete welfare ordering. It is a target-relative component used in economic choices that also contain ordinary payoff, feasibility, and continuation value.

For formal purposes, A^* is not modeled as a negotiated commitment device. Write $\mathbf{A}^* = (A_t^*)_{t \geq 0}$ for a predetermined, nonbinding target schedule. Each component A_t^* is fixed before the current action c_t is chosen, enters the recovery component only through the target-relative coordinate, and does not restrict the feasible action set. A treatment-reference schedule recorded before the current choice is one concrete interpretation, but the representation theorem assigns the target no independent normative status. The agent remains free to miss the target or to move away from it. All representation and application results are conditional on the given schedule $\mathbf{A}^*$. In the local calculations below, both the current target A^* and the applicable next-period target A^{**} are held fixed under perturbations of the current action and state. If the target instead responds endogenously to the state or action, the corresponding feedback derivatives must be added, and the stated comparative-static formulas need not remain unchanged.

Every continuous path that is monotone on a finite partition produces a four-module ledger: toward and away motion on the left and right of the target. The accessible ledger set is neither the full nonnegative cone nor a semigroup. Its geometry is constrained by the connectivity and endpoint flow of one path. Consequently, a standard additive-measurement argument cannot silently complete the accessible domain into an ordered semigroup and then cancel arbitrary formal sums. The same representation problem can arise in other target-relative path settings; the Becker–Murphy environment provides the principal economic interpretation and a disciplined state law.

This restricted domain makes location-free common-increment cancellation appear especially plausible. Its formal version, LC-A, permits every common formal increment whenever both translated ledgers remain accessible, and is therefore the strongest common-shift condition available without enlarging the primitive domain. Yet LC-A does not imply finite cancellation. The paper constructs an explicit strict order on the complete continuous path-generated ledger domain that satisfies strict directional monotonicity and LC-A but contains a balanced finite list of strict comparisons. The mechanism is specific to the partial domain: accessibility prevents arbitrary iteration of formal shifts, while a protected endpoint-flow state can survive every admissible common translation and act as a secondary tie breaker. Local translation tests therefore miss a global finite-list inconsistency.

The counterexample and the positive theorem play different logical roles. The counterexample rules out LC-A as a shortcut to additive consistency. The positive theorem then takes finite cancellation as an independent finite-list condition and gives a transparent sufficient route to representation; it is not claimed to provide a minimal axiomatization. Finite cancellation constructs a canonical partially ordered abelian group directly from actual comparisons and formal balance identities. Difference completeness and list Archimedean regularity provide sufficient conditions for a real-valued order embedding through Hölder’s theorem. A module-normalized path-generated monotone continuity condition then converts positive finitely additive interval contents into Stieltjes measures without excluding atoms.

The resulting representation has four modules:

$$U(\gamma) = \int h_{\gamma}^{T_L} d\lambda_{T_L} + \int h_{\gamma}^{T_R} d\lambda_{T_R} - \int h_{\gamma}^{A_L} d\lambda_{A_L} - \int h_{\gamma}^{A_R} d\lambda_{A_R}.$$

The measures may differ across direction and side. Under a directed-arrival convention, a target atom values each arrival at the target but does not create a separate first-arrival reward. When the left and right modules are compatible, the representation can be written with one toward and one away measure on the physical state space. In the absolutely continuous subclass it reduces to the toward/away density form used in the economic illustrations.

The addition application explains why target-relative direction can matter economically. In the smooth absolutely continuous subclass, directed recovery changes the full consumption-stock cross effect,

$$W_{cA} = u_{cA} + \eta R_d,$$

so the recovery wedge may strengthen, weaken, or reverse primitive adjacent complementarity. A separate one-step reduced-form problem shows that, when target pull dominates the remaining payoff variation, the maximizing response selects the feasible next state closest to the target. This projection geometry produces response plateaus and, when the same map is imposed repeatedly as a maintained transition rule, one-sided finite-time capture of sustainable positive targets. Sections 4 and 5 are illustrations of the representation, not a derivation of a Bellman-optimal rational-addiction policy.

The paper’s boundaries are explicit. It does not provide a complete welfare ordering, a joint empirical identification theorem for the target rule, coordinate, and four measures, or a model of withdrawal pain, cue-triggered hot states, or

dynamic inconsistency. Subjective targets may be formed and revised endogenously. Full accommodation, in which the target tracks the actual state, would eliminate the target-relative recovery gap and return the application to the Becker–Murphy benchmark. Modeling aspiration, revision, and commitment jointly requires additional primitives and is left for separate analysis. Difference completeness and list Archimedean regularity are sufficient real-valued regularity conditions rather than a claimed minimal characterization. Whether fixed-initial-state comparisons and economically observable bridge conditions can replace finite cancellation remains open.

Section 2 positions the paper in the rational-addiction, target-reference, path-preference, revealed-preference, and additive-measurement literatures. Section 3 is the main theoretical section: it defines continuous target-relative paths, characterizes ledger reachability, separates LC-A from finite cancellation, and proves the sufficient Stieltjes representation theorem. Sections 4 and 5 provide the recovery-wedge and target-projection illustrations. Appendix A synchronizes terminology and notation. Appendix B contains the representation proofs and the continuous-domain counterexample. Appendix C proves the illustrative application results.

2. Related literature

The leading economic application belongs to the rational-addiction literature. Becker and Murphy (1988) model addictive behavior as forward-looking dynamic optimization under stable preferences, with an addictive stock linking present and future consumption through adjacent complementarity, and Becker et al. (1994) bring that structure to empirical cigarette demand. Subsequent theories isolate mechanisms that can make quitting difficult even when the agent recognizes future harm. Orphanides and Zervos (1998) let past addictive consumption increase myopia; Gruber and Kőszegi (2001) embed time-inconsistent preferences and demand for self-control devices in the rational-addiction framework; Suranovic et al. (1999) explicitly model withdrawal and the decision to quit; and Bernheim and Rangel (2004) study cue-triggered decision processes. The present paper does not replace these mechanisms. It retains the Becker–Murphy stock dynamics and adds a different primitive: a stable preference component over target-relative directed motion. Recovery is therefore neither low consumption nor a terminal stock level, but the value assigned to the route by which the state approaches, crosses, or leaves a target.

The historical addiction literature also makes backsliding central. Winston (1980) models compulsive consumption as a conflict between wanting to consume and wanting not to consume, and discusses institutions that help agents purchase nonconsumption. Directed recovery does not reproduce that conflict or its institutional mechanism. Its narrower contribution is to retain the direction and multiplicity of passages in a conditional path preference, so that an improvement followed by backsliding need not be equivalent to a monotone improvement with the same endpoint.

The target interpretation is closest to the economic literature on goals, commitment, and reference points. Hsiaw (2013) and Koch and Nafziger (2011) show how self-chosen goals can serve as reference and self-regulation devices, while Giné et al. (2010) provide direct smoking-cessation evidence on a commitment contract. The target schedule used here is deliberately less structured than either a self-chosen goal or an enforceable contract: it has no menu effect, imposes no external penalty, and is not assigned independent welfare authority. It is a predetermined, nonbinding treatment reference. The paper conditions on that reference rather than explaining its endogenous formation or revision.

This distinction also separates directed recovery from neighboring preference models. Stigler and Becker (1977) emphasize stable preferences and shadow prices, while Pollak (1970) and Constantinides (1990) show how habit stocks and history-dependent benchmarks alter dynamic marginal incentives. Strotz (1955–1956), Laibson (1997), and O’Donoghue and Rabin (1999) study dynamic inconsistency, present bias, and commitment. By contrast, Gul and Pesendorfer (2001) axiomatize temptation and costly self-control through preferences over menus; their representative decision maker need not be present-biased or dynamically inconsistent. Kahneman and Tversky (1979) and Kőszegi and Rabin (2006) establish reference-dependent evaluation, while Kőszegi and Rabin (2009) develop a dynamic model of reference-dependent consumption plans in which news about present and future consumption changes utility. Simon (1955) supplies an earlier language of aspiration and acceptable states. Directed recovery takes neither a present-biased self, a menu temptation cost, nor an endogenous expectations-based reference point as primitive. Conditional on the predetermined target schedule, it asks when comparisons of repeated toward and away passages admit an additive path representation.

A neighboring path-preference literature studies dated consumption plans directly. Hindy et al. (1992) construct economically meaningful topologies and intertemporal preferences on continuous-time consumption plans. Their primitive retains the dating and timing of consumption. The present ledger instead records directed spatial passage

and multiplicity while deliberately abstracting from speed, dwell time, and the order of segments that generate the same ledger. The two approaches therefore address different notions of path dependence.

A second literature concerns revealed preference and finite consistency. Afriat (1967) and Varian (1982) show how finite choice data can be tested for compatibility with utility maximization. The finite-cancellation axiom used here has a related interpretation: a balanced finite list cannot make every left-hand path weakly better and one strictly better when both sides contain the same total directional content. The distinction is that the primitive domain here consists of path-generated ledgers rather than commodity bundles, and formal sums used in a balance certificate need not themselves be feasible single paths.

The mathematical core is closest to additive measurement. Luce and Tukey (1964), Krantz et al. (1971), and Fishburn (1970) study conditions under which ordinal comparisons admit additive numerical representations. Scott (1964) characterizes finite measurement structures through systems of linear inequalities and cancellation conditions. The present paper differs because its accessible ledger domain is only partially additive: it is generated by one-dimensional paths with endpoint and connectivity restrictions. The paper explicitly shows that strong local common-increment cancellation does not supply the missing finite cancellation.

The classical additive-measurement routes most relevant here formulate their cancellation requirements on product, concatenation, or finite comparison structures in which the combinations appearing in the axioms are themselves available alternatives, or they impose a global finite-cancellation condition directly. Neither route is automatic on $D = \mathcal{L}(\Gamma_F(X))$. A formal sum of two accessible ledgers need not be generated by any single continuous path, and the balanced sums used in finite cancellation are algebraic certificates rather than feasible experiences. Proposition 3.3 closes a specific possible shortcut: even LC-A, which allows every formal common increment whenever both translated ledgers happen to remain accessible, does not recover the missing finite-list condition. The ordered group in the positive theorem is therefore constructed from actual comparisons and balance identities rather than assumed by completing D into a semigroup.

Finite cancellation first yields a canonical partially ordered abelian group. Difference completeness and list Archimedean regularity make the relevant quotient a linearly ordered Archimedean group, and the final embedding step uses the classical Hölder theorem; a standard ordered-group reference is Fuchs (1963). This route is deliberately separated from the behavioral assumptions: the group is constructed as a conclusion from actual comparisons and balance identities, rather than assumed as a completion of an inaccessible ledger semigroup. The real-valued regularity conditions are presented as sufficient, not necessary.

The final representation step is measure-theoretic. Positive additive values on canonical half-open intervals define cumulative directional potentials. A path-generated monotone continuity condition gives the right continuity needed for Lebesgue–Stieltjes measures. Unlike Euclidean endpoint continuity or exact solvability, this condition only requires intervals decreasing to the empty set to become negligible and is therefore compatible with atoms. The target-point convention distinguishes repeated arrivals from a separate first-arrival event.

The two economic illustrations connect the representation back to addiction. The recovery wedge isolates how the directional transition term changes primitive BM adjacent complementarity. Target projection then converts strong recovery incentives into plateaus and capture regions. Clarke’s nonsmooth-analysis framework (Clarke, 1983) is used only at target points and constrained response boundaries.

3. Target-relative paths and directed recovery representation

This section separates the economic primitive from the algebra used to represent it. The primitive is a weak order over single continuous target-relative experiences. The algebra enters only after each experience has been mapped into its accessible directional ledger. This distinction is essential: accessible ledgers are generated by single paths and are generally not closed under addition.

3.1. States, targets, and the recovery coordinate

Let $A_t \in K$ be the state in period t , and let $c_t \in C$ be the current action. State dynamics are

$$A_{t+1} = F(A_t, c_t). \quad (3.1)$$

For the Becker–Murphy benchmark,

$$A_{t+1} = (1 - \delta)A_t + c_t, \quad 0 < \delta < 1. \quad (3.2)$$

Let $\mathbf{A}^* = (A_t^*)_{t \geq 0}$ be a predetermined, nonbinding target schedule, and let A_t^* be its period- t component fixed before the period- t action. The schedule is part of the conditioning environment: it does not remove any action from the feasible set, and the agent may fail to attain or may move away from the target. A reduction or treatment reference recorded before the current choice provides one concrete interpretation. What is fixed for the current choice is the applicable target schedule, not the future behavior of the state. The target-relative coordinate

$$x_t = \Psi(A_t, A_t^*) \quad (3.3)$$

takes values in a compact interval

$$X = [\underline{x}, \bar{x}], \quad \underline{x} < 0 < \bar{x},$$

and is normalized so that $x = 0$ is the target. Exit is the special case in which the target state is zero. A positive target instead describes reduction, maintenance, or a treatment plateau.

For the dynamic applications, the baseline coordinate is

$$\Psi(A, A^*) = \frac{A - A^*}{\chi + A + A^*}, \quad \chi > 0, \quad (3.4)$$

on nonnegative state and target spaces. It is strictly increasing in the actual state and strictly decreasing in the target. It is used as a bounded, smooth normalization rather than as a cardinal primitive. The representation theorem itself does not require this particular coordinate.

The recovery order introduced below is not a complete welfare order. It evaluates only target-relative directed motion. Its use in an actual choice problem is therefore conditional on the same complete initial state (A_0, A_0^*) and on the other payoff and continuation-value terms of that problem. The analysis also conditions on the target process. A subjective target could instead be selected and revised by the agent; in particular, full accommodation $A_t^* = A_t$ would make the target-relative coordinate and ledger vanish at each date. A model of target formation would require additional primitives governing aspiration, specification, and revision and is outside the present representation problem.

3.2. Continuous paths and directed arrivals

The primitive experience is a single continuous path

$$\Gamma_F(X) = \{\gamma : [0, T] \rightarrow X : \gamma \text{ is continuous and monotone on a finite partition}\}. \quad (3.5)$$

A path may reverse direction, cross the target, revisit an interval, and move on both sides of the target. The finite sequences used below are cutting points or turning points of this one experience; they are not independent disconnected segments.

Every path is cut at its turning points and at the endpoints of every maximal time interval on which it is at the target. Thus a path that crosses the target, touches it and returns, or remains there for a while is split into incoming, stationary, and outgoing pieces. Stationary pieces contribute the zero ledger. Each remaining nonconstant monotone segment belongs to exactly one of four modules,

$$\mathcal{M} = \{T_L, T_R, A_L, A_R\}, \quad (3.6)$$

where T denotes motion toward the target, A denotes motion away from the target, and L, R denote the physical side of the target.

It is convenient to give each module a coordinate that increases in the direction of the corresponding motion:

m	physical side	module coordinate y	Ξ_m
T_L	$x \leq 0$	$y = x$	$[x, 0]$
T_R	$x \geq 0$	$y = -x$	$[-\bar{x}, 0]$
A_L	$x \leq 0$	$y = -x$	$[0, -x]$
A_R	$x \geq 0$	$y = x$	$[0, \bar{x}]$

(3.7)

The ledger generated by a canonical module segment that moves from $y = a$ to $y = b > a$ is denoted $\tau_m(a, b) \in \mathcal{D}$. It has the single nonzero component $\mathbf{1}_{(a,b]}$ in module m .

We use a *directed-arrival convention*. Every module segment is recorded on a right-closed half-open interval $(a, b]$. Thus an incoming toward segment contains the target, whereas an outgoing away segment excludes it. Every arrival at the target—whether followed by a crossing, a return on the same side, or a stationary spell—ends an incoming toward segment. A subsequent departure begins an outgoing away segment. Consequently, the target is counted exactly once at each arrival and is not counted merely because a path starts there or remains there. A target atom therefore assigns value to every arrival at the target; it is not a reward for the first arrival only.

Definition 3.1 (Directional ledger). *For $\gamma \in \Gamma_F(X)$, let*

$$\ell(\gamma) = (\tilde{h}_\gamma^{T_L}, \tilde{h}_\gamma^{T_R}, \tilde{h}_\gamma^{A_L}, \tilde{h}_\gamma^{A_R}),$$

where $\tilde{h}_\gamma^m : \Xi_m \rightarrow \mathbb{N}$ is the sum of the canonical interval indicators contributed by all path segments in module m .

Each ledger component is a nonnegative integer-valued finite step function. Artificially cutting a monotone segment does not change the ledger because

$$\mathbf{1}_{(a,c]} = \mathbf{1}_{(a,b]} + \mathbf{1}_{(b,c]}, \quad a < b < c.$$

The same identity makes the ledger independent of the chosen finite monotone partition.

The geometry of the accessible domain can be stated independently of preferences. Let L be a nonnegative integer-valued finite-step quadruple on the four module intervals. Choose a physical partition

$$\underline{x} = z_0 < z_1 < \dots < z_N = \bar{x}$$

containing 0 and every physical breakpoint of L . On each adjacent band $[z_{j-1}, z_j]$, expand L into the number of upward and downward traversals: on the left, upward arcs are T_L and downward arcs are A_L ; on the right, upward arcs are A_R and downward arcs are T_R . Denote the resulting directed multigraph by $G(L, \mathcal{P})$.

Proposition 3.1 (Reachable finite-step ledgers). *A nonnegative integer-valued finite-step quadruple L belongs to $\mathcal{D}$ if and only if, for one (equivalently every common refinement) $\mathcal{P}$ as above, the non-isolated part of $G(L, \mathcal{P})$ is connected and there exist vertices s, t such that*

$$\deg^+(v) - \deg^-(v) = \mathbf{1}_{\{v=s\}} - \mathbf{1}_{\{v=t\}} \quad \text{for every vertex } v. \quad (3.8)$$

When $s = t$, all vertices are balanced. The vertices s, t are the initial and terminal states of a realizing path. Under the directed-arrival convention, the values of the toward components at the target equal the multiplicities of arcs entering 0 from the corresponding side.

The proposition is the directed Euler-trail characterization of a single connected experience. It makes the source of nonclosure explicit: adding two reachable ledgers combines two multigraphs, but the sum need not satisfy the endpoint balance and connected-trail conditions for one path. Appendix B.1 gives the proof.

3.3. The accessible ledger order

Let $\geq^P$ be the recovery order on $\Gamma_F(X)$.

Assumption 3.1 (Path weak order). *The relation $\geq^P$ is complete and transitive.*

Remark 3.1 (Structural order and observable fibers). *The weak order is a structural ranking of counterfactual experiences on the full path domain. Choice at a given date directly reveals only its restriction to paths with the same current state, target schedule, and feasible menu. The theorem does not claim that cross-initial-state comparisons are nonparametrically identified from such choices. Replacing the global order by fixed-initial-state orders linked through economically observable bridge conditions is a separate representation problem.*

Assumption 3.2 (Same-ledger indifference). *For all $\gamma, \eta \in \Gamma_F(X)$,*

$$\ell(\gamma) = \ell(\eta) \implies \gamma \sim^P \eta.$$

This is a substantive process restriction, not merely notation. It retains the side, direction, location, and multiplicity of passage, but abstracts from speed, dwell time, calendar duration, and the ordering of segments whenever these features leave the ledger unchanged. In particular, all constant paths have zero recovery ledger; stock-level utility is supplied by the ordinary payoff rather than by the recovery component.

Let

$$D = \ell(\Gamma_F(X)) \quad (3.9)$$

be the accessible ledger domain. Assumption 3.2 induces a well-defined order $\succeq$ on D :

$$L \succeq M \iff \gamma \succeq^P \eta$$

for any representatives with $\ell(\gamma) = L$ and $\ell(\eta) = M$. Write o for the zero ledger.

For a compact interval $\Xi = [\alpha, \beta]$, let $S_{\mathbb{Z}}^0(\Xi)$ be the abelian group generated by the indicators $\mathbf{1}_{(a,b]}$, $\alpha \leq a < b \leq \beta$. Its elements are integer-valued finite step functions that vanish at the lower endpoint α . Let

$$E = \bigoplus_{m \in \mathcal{M}} S_{\mathbb{Z}}^0(\Xi_m),$$

and let

$$F = \langle D - D \rangle_{\mathbb{Z}} \subseteq E. \quad (3.10)$$

Elements of F are finite formal differences of accessible ledgers. They need not themselves be ledgers of actual paths. Formal addition will be used to state balance conditions and to construct the representation; it does not enlarge the primitive preference domain.

Assumption 3.3 (Strict directional monotonicity). *For accessible ledgers $L, M \in \mathcal{D}$, suppose*

$$L^{TL} \geq M^{TL}, \quad L^{TR} \geq M^{TR}, \quad L^{AL} \leq M^{AL}, \quad L^{AR} \leq M^{AR}$$

pointwise, with at least one strict inequality on a set containing a nonempty interval. Then $L > M$.

This axiom compares only accessible ledgers. It does not assert that one may increase an arbitrary toward coordinate or delete an arbitrary away coordinate while remaining in $\mathcal{D}$.

3.4. Finite-list consistency and the boundary of common increments

Assumption 3.4 (Finite cancellation). *There do not exist accessible ledgers $L_i, M_i \in \mathcal{D}$, $i = 1, \dots, r$, such that*

$$L_i \geq M_i \text{ for every } i, \quad L_j > M_j \text{ for some } j,$$

and

$$\sum_{i=1}^r L_i = \sum_{i=1}^r M_i \text{ in } E. \quad (3.11)$$

Finite cancellation is a finite-list consistency condition on actual path comparisons. The equality in (3.11) says that the two sides contain exactly the same total directional content. It does not say that either formal sum is itself an accessible single-path ledger.

For comparison with more local conditions, define location-free cancellation for common increments. LC-A permits any nonnegative formal step-function ledger K for which both translated ledgers are accessible:

$$K \in E_+, \quad L, M, L + K, M + K \in \mathcal{D} \implies [L \geq M \iff L + K \geq M + K]. \quad (3.12)$$

LC-P is the weaker version that additionally requires $K \in \mathcal{D}$.

Proposition 3.2 (Finite cancellation implies common-increment cancellation). *Under a weak order, Assumption 3.4 implies LC-A and hence LC-P.*

Proof. Suppose $L \geq M$ but $M + K > L + K$. The two comparisons form a balanced strict list because

$$L + (M + K) = M + (L + K),$$

contradicting finite cancellation. This proves one direction of (3.12); the reverse direction is symmetric. $\square$

The converse fails even on the full, untruncated path domain.

Proposition 3.3 (Common increments do not imply finite cancellation). *On the full accessible ledger domain generated by $\Gamma_F(X)$, there exists a strict total order satisfying strict directional monotonicity and LC-A but violating finite cancellation. Its pullback through ℓ is a path weak order satisfying same-ledger indifference.*

The proof is in Appendix B. It constructs a translation-invariant primary ranking of the directional ledger and uses the net endpoint-flow state to break ties that no admissible common increment can alter. On two adjacent bands on one side of the target, the global order contains the balanced strict certificate

$$\begin{aligned} (1, 1, 0, 0) &> (2, 2, 3, 3), \\ (0, 2, 0, 3) &> (0, 1, 0, 0), \\ (2, 0, 3, 0) &> (1, 0, 0, 0), \end{aligned} \tag{3.13}$$

where coordinates are (T_1, T_2, A_1, A_2) . The two sides of the three comparisons both sum to $(3, 3, 3, 3)$. Thus LC is a necessary two-comparison test implied by finite cancellation, but it is not a substitute for finite-list consistency.

3.5. Difference comparisons and real-valued regularity

Finite cancellation gives a canonical additive representation in a partially ordered group. To obtain a single real-valued representation through a transparent route, we impose two additional conditions on finite lists.

Let $\mathfrak{L}(D)$ be the collection of finite lists of accessible ledgers, including the empty list. For

$$\mathbf{A} = (A_1, \dots, A_r), \quad \mathbf{B} = (B_1, \dots, B_s),$$

define

$$\Delta(\mathbf{A}, \mathbf{B}) = \sum_{i=1}^r A_i - \sum_{j=1}^s B_j \in F.$$

Write

$$\mathbf{A} \geq_{\Delta} \mathbf{B}$$

if there exist finitely many actual weak comparisons $L_k \geq M_k$, $k = 1, \dots, q$, such that

$$\sum_i A_i + \sum_k M_k = \sum_j B_j + \sum_k L_k. \tag{3.14}$$

Thus $\mathbf{A} \geq_{\Delta} \mathbf{B}$ means that the formal difference $\Delta(\mathbf{A}, \mathbf{B})$ is generated by a finite list of observed weak improvements. Define $>_{\Delta}$ as the asymmetric part.

Assumption 3.5 (Difference completeness). *For all finite accessible-ledger lists $\mathbf{A}, \mathbf{B}$,*

$$\mathbf{A} \geq_{\Delta} \mathbf{B} \quad \text{or} \quad \mathbf{B} \geq_{\Delta} \mathbf{A}.$$

Assumption 3.6 (List Archimedean regularity). *If*

$$\mathbf{A} >_{\Delta} \mathbf{B} \quad \text{and} \quad \mathbf{C} >_{\Delta} \mathbf{D},$$

then there is an integer $n \geq 1$ such that

$$(n\mathbf{A}) \sqcup \mathbf{D} \geq_{\Delta} (n\mathbf{B}) \sqcup \mathbf{C}. \tag{3.15}$$

Here $n\mathbf{A}$ repeats the list $\mathbf{A}$ n times, and $\sqcup$ denotes list concatenation.

Difference completeness requires every finite formal difference generated by actual ledgers to have a revealed sign. List Archimedean regularity rules out a positive difference that remains infinitesimal relative to another positive difference under every finite repetition. Both conditions are stated using actual ledgers, actual weak comparisons, and finite balance identities.

Remark 3.2 (Algebraic role and empirical content of DC and LA). *Difference completeness and list Archimedean regularity are algebraic extension conditions stated through finite certificates of actual ledger comparisons. DC requires every formal difference generated by accessible ledgers to have a sign in the canonical extension; LA excludes lexicographic or infinitesimal positive differences. They are stronger than completeness and continuity of the primitive path order and are not presented as directly observable one-menu axioms. A finite dataset can refute DC or LA by exhibiting incompatible comparison certificates, but confirmation on a finite dataset cannot establish their unrestricted versions. Their role is to provide a transparent sufficient extension theorem from the canonical ordered group to $\mathbb{R}$; they are not claimed to be necessary, weakest, or part of a necessary-and-sufficient behavioral characterization.*

Example 3.1 (A path-sensitive atomic class satisfying all assumptions). *Let $X = [-4, 4]$ and let $\rho_L, \rho_R \in [0, 2]$. Put a side-specific atom at the target endpoint of each toward module and define*

$$\begin{aligned} \hat{U}_{\rho_L, \rho_R}(L) = & \int_{\Xi_{T_L}} L^{T_L}(y) (dy + \rho_L \delta_0(dy)) + \int_{\Xi_{T_R}} L^{T_R}(y) (dy + \rho_R \delta_0(dy)) \\ & - 2 \int_{\Xi_{A_L}} L^{A_L}(y) dy - 2 \int_{\Xi_{A_R}} L^{A_R}(y) dy. \end{aligned} \quad (3.16)$$

Order accessible ledgers by the numerical value of $\hat{U}_{\rho_L, \rho_R}$ and pull the order back to the full path domain through the ledger map. The resulting path order satisfies same-ledger indifference, strict directional monotonicity, finite cancellation, difference completeness, list Archimedean regularity, and PGM C^+ . If $\rho_L = \rho_R$, the four module measures satisfy target-atom compatibility and can be stitched into the two-measure form. If $\rho_L \neq \rho_R$, the four-module representation still holds, but no single physical toward measure can preserve both arrival values.

The class is genuinely path-sensitive. Inserting a closed excursion of physical length $r > 0$ in an interval that does not touch the target changes value by $r - 2r = -r < 0$, so paths with the same endpoints can be ranked differently. The case $\rho_L = \rho_R = 0$ gives the atomless benchmark. Appendix B.4 gives the complete verification.

Proposition 3.4 (Canonical group-valued representation). *Under a weak order and finite cancellation, there exist a partially ordered abelian group G and a map $u_0 : \mathcal{D} \rightarrow G$ such that*

$$L \geq M \iff u_0(L) \geq u_0(M).$$

Moreover, every balance identity in E is preserved:

$$\sum_i L_i = \sum_j M_j \implies \sum_i u_0(L_i) = \sum_j u_0(M_j).$$

The construction is canonical in the factorization sense stated in Appendix B.

Proposition 3.5 (Real-valued additive representation). *Under a weak order and Assumptions 3.4–3.6, there exists an exact real-valued ledger representation $\hat{U} : \mathcal{D} \rightarrow \mathbb{R}$ such that*

$$L \geq M \iff \hat{U}(L) \geq \hat{U}(M),$$

and $\hat{U}$ extends to a group homomorphism

$$\tilde{U} : F \rightarrow (\mathbb{R}, +).$$

If the order is nontrivial, any two exact real-valued ledger representations that extend to group homomorphisms on F differ by a common positive multiplicative factor. Equivalently, every such representation factors through the canonical quotient group G , and the induced map on G is an order embedding.

The proof first constructs the quotient ordered group generated by finite weak-comparison differences. Difference completeness makes its order linear, list Archimedean regularity makes it Archimedean, and Hölder's theorem then embeds it as an ordered subgroup of $(\mathbb{R}, +)$.

3.6. Module-normalized path continuity

For each module, define the order in which a larger canonical interval means a larger amount of the positively normalized directional content:

$$A \succeq_m^+ B \iff \begin{cases} A \geq B, & m \in \{T_L, T_R\}, \\ B \geq A, & m \in \{A_L, A_R\}. \end{cases}$$

The reversal for away modules is necessary because more away motion lowers recovery value.

Assumption 3.7 (Path-generated monotone continuity). *Fix a module m . For every sequence $b_n \downarrow a$ with $b_n > a$, and every actual canonical interval ledger $J = \tau_m(c, d)$ in the same module,*

$$\tau_m(a, b_n) \succeq_m^+ J \text{ for every } n \implies o \succeq_m^+ J. \quad (3.17)$$

This condition is imposed only on actual connected canonical paths. It does not define preferences over disconnected sets, does not refer to a previously constructed numerical content, and does not require the value of intervals shrinking to a singleton to vanish. It therefore permits Stieltjes atoms.

3.7. Directed Stieltjes representation

Theorem 3.1 (Directed Stieltjes representation). *Suppose the path order satisfies Assumptions 3.1–3.2. Suppose its induced accessible-ledger order satisfies strict directional monotonicity, finite cancellation, difference completeness, list Archimedean regularity, and path-generated monotone continuity in every module. The full path domain $\Gamma_F(X)$ contains every canonical module interval and every finite subdivision of such an interval. Then there exist an exact ledger representation $\hat{U}$, the exact path representation*

$$U(\gamma) := \hat{U}(\ell(\gamma)),$$

and four finite nonnegative Borel measures

$$\lambda_{T_L}, \quad \lambda_{T_R}, \quad \lambda_{A_L}, \quad \lambda_{A_R},$$

defined on their respective module intervals $\Xi_m = [\alpha_m, \beta_m]$, normalized by

$$\lambda_m(\{\alpha_m\}) = 0, \quad m \in \mathcal{M},$$

such that every $\gamma \in \Gamma_F(X)$ satisfies

$$\begin{aligned} U(\gamma) = & \int_{\Xi_{T_L}} \tilde{h}_\gamma^{T_L}(y) \lambda_{T_L}(dy) + \int_{\Xi_{T_R}} \tilde{h}_\gamma^{T_R}(y) \lambda_{T_R}(dy) \\ & - \int_{\Xi_{A_L}} \tilde{h}_\gamma^{A_L}(y) \lambda_{A_L}(dy) - \int_{\Xi_{A_R}} \tilde{h}_\gamma^{A_R}(y) \lambda_{A_R}(dy). \end{aligned} \quad (3.18)$$

The measures may have atoms away from their normalized lower endpoints. Under the directed-arrival convention, a toward atom at the target is counted once for every incoming arrival and never as a separate first-arrival event; the away target atoms are zero because the target is their lower endpoint. For a fixed normalization of $\hat{U}$, each λ_m is unique. If the order is nontrivial, the joint representation is unique up to a common positive multiplicative factor within the exact additive class stated in Proposition 3.5.

Proof outline. Proposition 3.5 supplies the ledger representation $\hat{U}$ and its group-homomorphic extension $\tilde{U}$. For a canonical interval in module m , define the positively normalized interval content

$$v_m((a, b]) = \varepsilon_m \hat{U}(\tau_m(a, b)), \quad \varepsilon_m = \begin{cases} 1, & m \in \{T_L, T_R\}, \\ -1, & m \in \{A_L, A_R\}. \end{cases}$$

Strict directional monotonicity makes v_m positive, and the formal ledger identity

$$\tau_m(a, c) = \tau_m(a, b) + \tau_m(b, c)$$

gives finite additivity. Assumption 3.7, together with arbitrary finite subdivision of a real interval, implies

$$b_n \downarrow a \implies v_m((a, b_n]) \rightarrow 0.$$

Hence

$$G_m(x) = v_m((\alpha_m, x])$$

is monotone and right-continuous. The Lebesgue–Stieltjes construction, with $\lambda_m(\{\alpha_m\}) = 0$, gives a unique finite Borel measure satisfying

$$\lambda_m((a, b]) = v_m((a, b]).$$

Lemma B.7 shows that every path-ledger component is a finite sum of canonical interval indicators, yielding (3.18). Full details, including the target endpoint audit and uniqueness, are in Appendix B. $\square$

The four-module form is the general result. It permits recovery from the left and recovery from the right to carry different target atoms and different position weights.

Corollary 3.1 (Side-compatible two-measure form). *Push each λ_m forward to its physical half-interval and denote the result by μ_m . If the two toward measures assign the same mass to the target,*

$$\mu_{T_L}(\{0\}) = \mu_{T_R}(\{0\}), \quad (3.19)$$

then the left and right measures can be stitched into finite nonnegative Borel measures μ^T and μ^A on X such that

$$U(\gamma) = \int_X h_\gamma^T(z) \mu^T(dz) - \int_X h_\gamma^A(z) \mu^A(dz). \quad (3.20)$$

The common target atom is inserted once in μ^T ; it is not obtained by adding two zero-extended target atoms. Under the directed-arrival convention, $\mu^A(\{0\}) = 0$.

Remark 3.3 (Two boundaries of target atoms). *Target atoms are compatible with PGM C^+ , but they create two distinct boundaries. First, if the left and right toward target atoms differ, the four-module representation remains valid while the two-measure form generally fails: one physical toward measure can assign only one mass to the target. Second, if a toward target atom on one side has mass $\rho > 0$ and feasible paths permit arbitrarily short closed excursions that touch the target and immediately return, then sufficiently short excursions have positive value. The incoming segment retains the fixed atom ρ , whereas the finite away measure of the shrinking return interval converges to zero. A target atom therefore does not by itself satisfy a recursive no-positive-loop discipline. Recursive applications must exclude arbitrarily short touch loops, add a separate loop cost, or work in the atomless absolutely continuous subclass. The recovery-wedge and target-projection results below use the absolutely continuous subclass.*

Corollary 3.2 (No-positive-loop discipline excludes target atoms). *Suppose a side admits arbitrarily short touch-and-return excursions and every closed excursion on that side is weakly nonpositive. Under the directed-arrival convention and finite away measure, the toward measure assigns zero mass to the target on that side.*

Proof. A positive target atom would generate a strictly positive sufficiently short excursion by Proposition B.1, contradicting the no-positive-loop condition. $\square$

Corollary 3.3 (Absolutely continuous subclass). *If the four physical module measures are absolutely continuous with respect to Lebesgue measure, then their left and right densities can be joined piecewise into nonnegative functions q^T, q^A on X such that*

$$U(\gamma) = \int_X q^T(z) h_\gamma^T(z) dz - \int_X q^A(z) h_\gamma^A(z) dz. \quad (3.21)$$

No additional target-atom compatibility condition is needed in this subclass, because absolute continuity makes all singleton masses zero. The smooth subclass used in the subsequent dynamic analysis imposes the additional local differentiability and lower-bound conditions stated there.

Proposition 3.6 (Coordinate covariance of the representation). *For each module m , let $\phi_m : \Xi_m \rightarrow \bar{\Xi}_m$ be a strictly increasing homeomorphism, and define*

$$\bar{h}_\gamma^m(\bar{y}) = \tilde{h}_\gamma^m(\phi_m^{-1}(\bar{y})), \quad \bar{\lambda}_m = (\phi_m)_\# \lambda_m.$$

Then

$$\int_{\Xi_m} \tilde{h}_\gamma^m(y) \lambda_m(dy) = \int_{\bar{\Xi}_m} \bar{h}_\gamma^m(\bar{y}) \bar{\lambda}_m(d\bar{y}) \quad (3.22)$$

for every path and every module. Hence the four-module value and the induced path order are invariant to monotone reparameterization of the module coordinates. A strictly increasing physical reparameterization $\tilde{x} = \phi(x)$ with $\phi(0) = 0$ is the corresponding special case and preserves the directed-arrival convention and target atoms.

If ϕ_m is a C^1 diffeomorphism and λ_m has density q_m , then $\bar{\lambda}_m$ has density

$$\bar{q}_m(\bar{y}) = \frac{q_m(\phi_m^{-1}(\bar{y}))}{\phi'_m(\phi_m^{-1}(\bar{y}))}. \quad (3.23)$$

Thus densities and their slopes depend on the coordinate normalization even though total path values do not.

Proof. Equation (3.22) is the change-of-variables identity for a pushforward measure. The density formula is the ordinary Radon–Nikodym change of variables. Because each ϕ_m is increasing, it preserves interval order and the half-open endpoint convention; if $\phi(0) = 0$, a target atom is pushed to the same distinguished target point. $\square$

For later use, let $\mathfrak{D}^{\text{reg}}$ denote the class covered by Theorem 3.1, let $\mathfrak{D}^{\text{ac}}$ be its absolutely continuous subclass in Corollary 3.3, and let $\mathfrak{D}^{\text{smooth}}$ be the local subclass in which the relevant densities, the recovery coordinate, and response branches have the differentiability required below. Thus

$$\mathfrak{D}^{\text{smooth}} \subset \mathfrak{D}^{\text{ac}} \subset \mathfrak{D}^{\text{reg}}. \quad (3.24)$$

3.8. One-period transitions and interpretation

For a one-period comparison from x to x' , let $\gamma_{x,x'}$ be the canonical monotone path from x to x' , split at the target if necessary. The path utility is $U = \hat{U} \circ \ell$. Define the target-relative transition value

$$T(x, x') = U(\gamma_{x,x'}). \quad (3.25)$$

On the side-compatible absolutely continuous subclass,

$$T(x, x') = \int_X q^T(z) h_{x,x'}^T(z) dz - \int_X q^A(z) h_{x,x'}^A(z) dz. \quad (3.26)$$

For notational continuity with the dynamic sections, one may write $q^+ = q^T$, $q^- = q^A$, $h^+ = h^T$, and $h^- = h^A$ in this subclass.

The representation distinguishes three objects that terminal distance alone does not: the direction of motion, the positions through which motion passes, and the number of repeated passages. It also clarifies the empirical meaning of the axioms. LC checks invariance to one common increment, whereas finite cancellation checks consistency of every finite balanced comparison list. Difference completeness and list Archimedean regularity then provide a sufficient route to a common cardinal scale. Finally, PGMC is the path-domain continuity condition that converts finite interval content into Stieltjes measures without excluding atoms.

The representation deliberately abstracts from speed, calendar duration, and the temporal ordering of segments that generate the same ledger. Every constant path has the zero recovery ledger, irrespective of the state at which it is constant. Thus T is a process value for directed passage, not a utility index for the stock level itself; ordinary payoff and continuation value distinguish stationary states. In this precise sense the model is sensitive to direction, location, and multiplicity, but not to every feature of a dated narrative.

The recovery ranking remains a partial component of economic evaluation. The one-step comparisons in Sections 4–5 combine $T(x, x')$ with ordinary current payoff, feasibility, and continuation value. In those applications, the treatment reference or target schedule is specified before the relevant consumption choice and is therefore held fixed when comparing feasible actions. This timing statement is interpretive; it adds no constraint to the feasible set and no axiom to the path order. The open problem of deriving finite cancellation from preferences observed only within fixed-initial-state fibers is separate from the global representation theorem above.

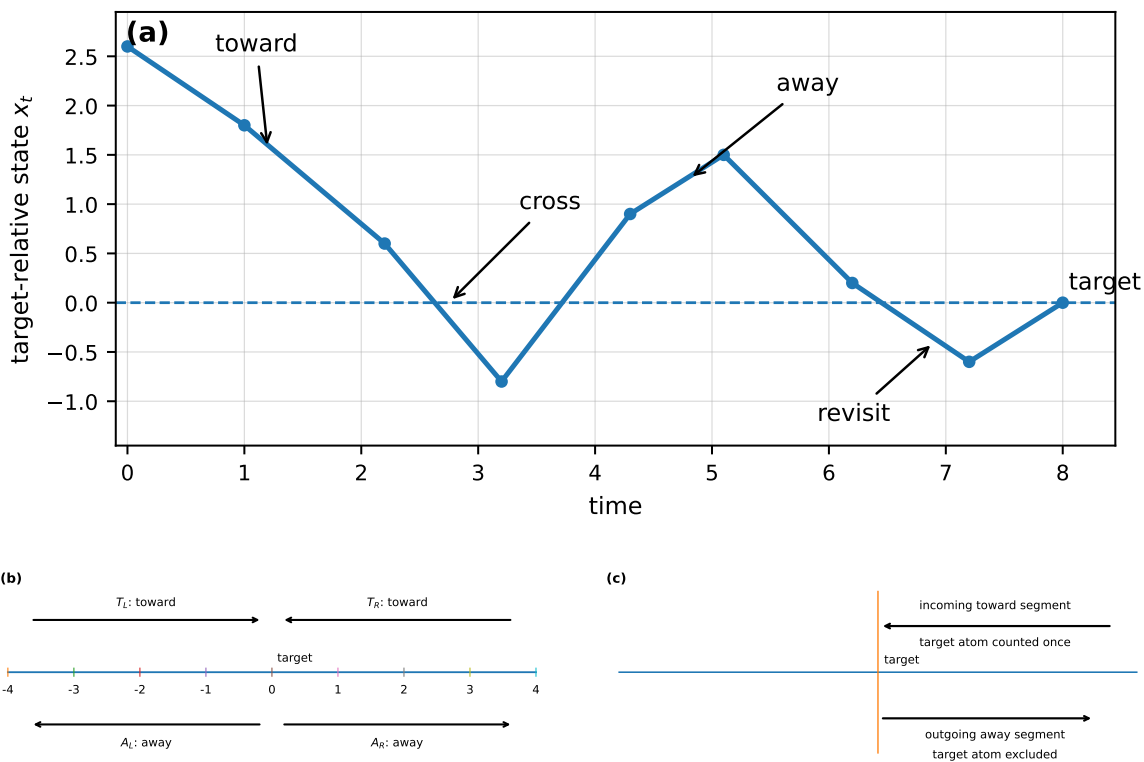


Figure 1: Target-relative path accounting. Panel (a) shows a single continuous path that approaches, crosses, leaves, and revisits the target. Panel (b) assigns each nonstationary monotone segment to one of four directional modules. Panel (c) shows the directed-arrival convention: an incoming toward segment includes the target, while the outgoing away segment excludes it.

4. Illustrative One-Step Marginal Structure: The Recovery Wedge

This section gives the first illustrative economic implication of the representation. The projection plateau in Section 5 rests on a more basic local force: directed recovery changes the cross marginal between current consumption and the state stock. The primitive payoff $u(c, A)$ retains the Becker–Murphy adjacent complementarity u_{cA} . Once the recovery term generated by the target-relative transition enters the payoff, the full local cross effect is determined jointly by the primitive term and the transition term. The difference between the two is the recovery wedge.

The Stieltjes kernels in Section 3 provide the representation layer. To derive local marginal formulas, this section works in an absolutely continuous and smooth subclass of the position-kernel representation. In the direction-fixed neighborhood under study, the active directional kernel satisfies

$$\mu^d \ll \lambda, \quad q^d = \frac{d\mu^d}{d\lambda}, \quad q^d \in C^1,$$

where $d \in \{+, -\}$ is the locally active direction. The recovery term can then be written as an endpoint integral and has the local first and second derivatives used below.

The following object is a one-step reduced-form payoff, not a Bellman envelope. The one-period full payoff is

$$W(c, A) = u(c, A) + \eta T(x, x'), \quad (4.1)$$

where

$$x = \Psi(A, A^*), \quad x' = \Psi(A', A^{*'}).$$

Here $(A^*, A^{*'})$ is the portion of the treatment target schedule applicable to the current transition and fixed before c is chosen. It is therefore predetermined in the local derivative calculation, while consumption remains unrestricted within its feasible set. The recovery term $T(x, x')$ is determined by the directed motion of the target-relative state. Hence the full cross effect is

$$W_{cA} = u_{cA} + \eta T_{cA}.$$

4.1. Local environment

Consider the linear BM state law

$$A' = (1 - \delta)A + c, \quad 0 < \delta < 1. \quad (4.2)$$

In this section, the target schedule is held fixed under both local perturbations used in the cross derivative; equivalently, $\partial A^{*'} / \partial c = \partial A^{*'} / \partial A = 0$. The formula is conditional on this schedule. Define

$$x = \Psi(A, A^*), \quad x' = p(A') = \Psi(A', A^{*'}). \quad (4.3)$$

Consider a local region in which the direction is fixed:

$$xx' > 0, \quad x' \neq x.$$

The first condition says that the current transition does not cross the target; the second says that there is nonzero target-relative motion.

Let

$$s = \text{sgn}(x').$$

If $s = 1$, the state is above the target; if $s = -1$, it is below the target. Let $d \in \{+, -\}$ denote the active direction in the local region:

$$d = + \iff |x'| < |x|,$$

$$d = - \iff |x'| > |x|.$$

Thus $d = +$ denotes motion toward the target and $d = -$ denotes motion away from the target. In this local region, the density representation in Corollary 3.3 reduces to an endpoint integral. The one-period recovery term is represented by the active directional density q^d , and

$$T_{x'}(x, x') = -sq^d(x'), \quad T_{xx'}(x, x') = 0.$$

The equality $T_{xx'} = 0$ follows from the path-additive endpoint-integral representation. If one allowed nonadditive $T(x, x')$, the wedge formula would contain an additional term $T_{xx'}x_Ax'_c$.

4.2. The local cross effect and the recovery wedge

Proposition 4.1 (Recovery wedge in the local cross effect). *This proposition works in a direction-fixed local region of $\mathfrak{D}^{\text{smooth}}$. Suppose*

$$A' = (1 - \delta)A + c, \quad 0 < \delta < 1,$$

and

$$x' = p(A') = \Psi(A', A^{*'}).$$

In the local region

$$xx' > 0, \quad x' \neq x,$$

let

$$s = \text{sgn}(x'),$$

and let $d \in \{+, -\}$ be the active direction. If $q^d \in C^1$ and $p \in C^2$, then the full payoff

$$W(c, A) = u(c, A) + \eta T(x, x')$$

satisfies

$$W_{cA} = u_{cA} - \eta s(1 - \delta) \left[q^{d'}(x')(p'(A'))^2 + q^d(x')p''(A') \right]. \quad (4.4)$$

Define

$$\mathcal{R}_d(c, A) = -s(1 - \delta) \left[q^{d'}(x')(p'(A'))^2 + q^d(x')p''(A') \right]. \quad (4.5)$$

Then

$$W_{cA} = u_{cA} + \eta \mathcal{R}_d. \quad (4.6)$$

The term $\mathcal{R}_d$ is the directed recovery wedge.

Proof. In a direction-fixed local region, the marginal recovery value with respect to the endpoint x' is

$$T_{x'} = -sq^d(x').$$

Therefore

$$T_c = T_{x'}x'_c = -sq^d(x')x'_c.$$

Since

$$x' = p(A'), \quad A' = (1 - \delta)A + c,$$

we have

$$x'_c = p'(A'), \quad x'_A = (1 - \delta)p'(A'), \quad x'_{cA} = (1 - \delta)p''(A').$$

Hence

$$T_{cA} = -s \left[q^{d'}(x')x'_Ax'_c + q^d(x')x'_{cA} \right].$$

Substituting the three derivatives of x' gives

$$T_{cA} = -s(1 - \delta) \left[q^{d'}(x')(p'(A'))^2 + q^d(x')p''(A') \right].$$

Since $W_{cA} = u_{cA} + \eta T_{cA}$, the result follows. □

Remark 4.1 (Target feedback). *If the applicable target responds to the current state, for example $A^{*'} = G(A)$, then x'_A and x'_{cA} contain derivatives of G and the chain rule adds target-feedback terms to (4.4). Proposition 4.1 therefore identifies the recovery wedge conditional on a predetermined target schedule; it is not a formula for an endogenously revised target without further primitives.*

Corollary 4.1 (Directed recovery can reverse local BM complementarity). *Under the conditions of Proposition 4.1, suppose the primitive payoff satisfies local adjacent complementarity, $u_{cA} > 0$. If $\mathcal{R}_d < 0$ at a point and*

$$\eta > \frac{u_{cA}}{-\mathcal{R}_d},$$

then the full payoff satisfies

$$W_{cA} < 0.$$

Thus a stock that is locally complementary to consumption in the Becker–Murphy primitive payoff can become a local consumption-discouraging state in the full model when the directed recovery motive is sufficiently strong. The sign of effective adjacent complementarity is determined by the joint force of primitive complementarity and the recovery wedge.

Proof. This follows directly from $W_{cA} = u_{cA} + \eta \mathcal{R}_d$. □

Proposition 4.1 and equations (4.4)–(4.6) show that directed recovery modifies BM adjacent complementarity through two structural terms. The first, the slope $q^{d'}(x')$ of the directional kernel, records how marginal recovery value changes with target-relative position. The second, the curvature $p''(A')$ of the target-relative coordinate, records how the same change in the actual stock has different economic meanings at different target-relative positions. The recovery wedge is therefore generated jointly by directional-kernel slopes and target-coordinate curvature. Equivalently, u_{cA} measures primitive complementarity in the baseline payoff, while $\mathcal{R}_d$ measures how target-relative motion transforms that primitive complementarity into effective complementarity in the full model.

Remark 4.2 (Coordinate invariance of the recovery wedge). *Let $\tilde{x} = \phi(x)$ for a strictly increasing C^2 diffeomorphism with $\phi(0) = 0$, let $\tilde{p} = \phi \circ p$, and transform the active density according to Proposition 3.6. Then*

$$\tilde{q}^{d'}(\tilde{p})(\tilde{p}')^2 + \tilde{q}^d(\tilde{p})\tilde{p}'' = q^{d'}(p)(p')^2 + q^d(p)p''. \quad (4.7)$$

Consequently, the transition value T , the composite derivative T_{cA} , and the total wedge $\mathcal{R}_d$ are coordinate invariant. The separate labels “kernel-slope effect” and “coordinate-curvature effect” are normalization-dependent components of that invariant sum and should not be interpreted separately without fixing a coordinate convention.

Remark 4.3 (Direction switches and nonsmooth boundaries). *Proposition 4.1 is not asserted at $x = 0$, $x' = 0$, or a direction switch. On each adjacent smooth branch, the corresponding one-sided cross derivative is the limit of (4.4). If the composite payoff is locally Lipschitz, its Clarke generalized cross marginal is contained in the convex hull of the finite branch limits. A single smooth equality at the boundary requires those limits to coincide and is not assumed here.*

4.3. The recovery wedge under the baseline coordinate

Take the baseline target-relative coordinate

$$p(A') = \frac{A' - A^{*'}}{\chi + A' + A^{*'}}, \quad \chi > 0. \quad (4.8)$$

Then

$$p'(A') = \frac{\chi + 2A^{*'}}{(\chi + A' + A^{*'})^2},$$

$$p''(A') = -\frac{2(\chi + 2A^{*'})}{(\chi + A' + A^{*'})^3}.$$

Substitution into Proposition 4.1 gives

$$W_{cA} = u_{cA} + \eta s(1 - \delta) \frac{\chi + 2A^{*'}}{(\chi + A' + A^{*'})^4} \left[2q^d(x')(\chi + A' + A^{*'}) - q^{d'}(x')(\chi + 2A^{*'}) \right]. \quad (4.9)$$

In this expression,

$$2q^d(x')(\chi + A' + A^{*'})$$

comes from the curvature of the target-relative coordinate, while

$$q^{d'}(x')(\chi + 2A^{*'})$$

comes from the slope of the directional kernel.

4.4. Reinforcement, weakening, and reversal of adjacent complementarity

By the decomposition (4.6), if $\mathcal{R}_d > 0$, directed recovery reinforces BM adjacent complementarity:

$$W_{cA} > u_{cA} > 0.$$

If $\mathcal{R}_d < 0$ and

$$u_{cA} + \eta \mathcal{R}_d > 0,$$

directed recovery weakens BM adjacent complementarity while the full payoff remains locally complementary. If

$$\eta |\mathcal{R}_d| > u_{cA} \quad \text{and} \quad \mathcal{R}_d < 0,$$

then

$$W_{cA} < 0.$$

In this case, the primitive BM payoff preserves adjacent complementarity between consumption and the addictive stock, but the local relation in the full payoff becomes substitutive.

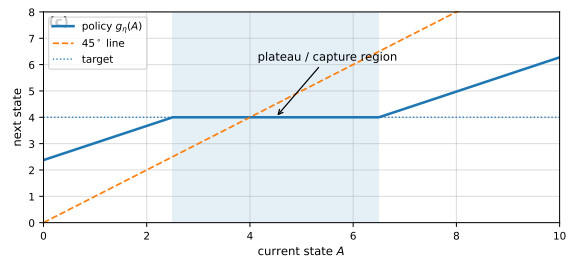
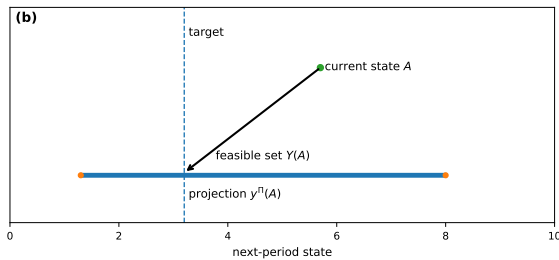
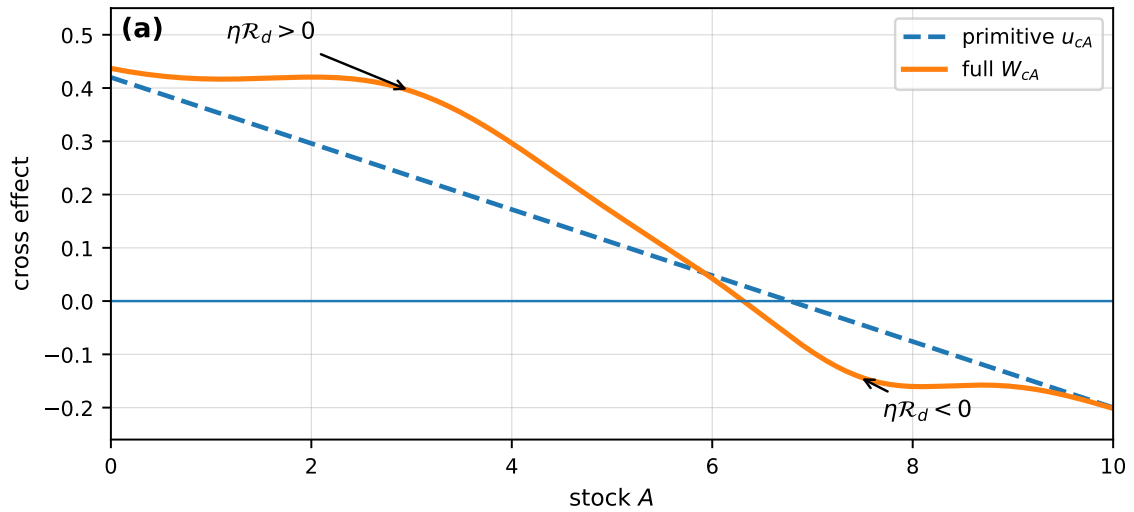


Figure 2: Economic implications of directed recovery. Panel (a) illustrates how the recovery wedge shifts the full cross effect relative to the primitive BM term and can change its sign. Panel (b) shows target projection on a feasible next-state interval. Panel (c) shows the resulting projection plateau and capture region when the target is reachable. The curves are schematic and are not a calibration.

5. Illustrative Target Geometry: One-Step Projection and Conditional Capture

Section 4 derived a conditional one-step marginal force. This section studies the corresponding one-step maximization geometry. Target projection identifies the feasible endpoint selected when the directed-recovery term dominates an η -independent reduced-form residual. The resulting projection plateau is an implication of that one-step problem. Repeated-projection capture is then a conditional geometric statement obtained by iterating the same map; it is not derived here as the optimal policy of an infinite-horizon Bellman problem.

The section works in a local absolutely continuous subclass: the relevant target-relative intervals have directional densities with positive lower bounds. The chain is

$$\text{one-step recovery dominance} \longrightarrow \text{target projection} \longrightarrow \text{projection plateau} \longrightarrow \text{conditional capture}.$$

Remark 5.1 (Positive interpretation and scope of projection). *The projection statements are positive properties of the stated one-step objective. The target is a predetermined, nonbinding reference and does not remove any feasible consumption choice. The results do not rank captured paths normatively, do not identify a welfare-prioritized “true self,” and do not establish a Bellman-optimal policy. Those interpretations require additional normative or recursive structure.*

Consider the standard linear BM dynamics

$$A^+ = (1 - \delta)A + c, \quad c \in [0, \bar{c}]. \quad (5.1)$$

Given A , choosing consumption c is equivalent to choosing the next-period state $y = A^+$. The consumption that implements y is uniquely

$$c_A(y) = y - (1 - \delta)A.$$

5.1. Feasible sets and target projection

Given the current state A , the next-period feasible set is

$$Y(A) = \{(1 - \delta)A + c : c \in [0, \bar{c}]\}, \quad (5.2)$$

that is,

$$Y(A) = [(1 - \delta)A, (1 - \delta)A + \bar{c}]. \quad (5.3)$$

Let the next-period target be A^{*+} and define

$$\xi(y) = \Psi(y, A^{*+}).$$

The target-projection set is

$$\boxed{\Pi_A = \arg \min_{y \in Y(A)} |\xi(y)|.} \quad (5.4)$$

When the projection is unique, denote its element by $y^\Pi(A)$. Under the baseline coordinate, ξ is strictly increasing in y , so the projection is unique. If $A^{*+} \in Y(A)$, then

$$y^\Pi(A) = A^{*+}.$$

Collect all terms other than the current recovery term in

$$B_A : Y(A) \rightarrow \mathbb{R}. \quad (5.5)$$

Here B_A is an η -independent reduced-form residual for the one-step problem. It may summarize primitive current payoff and a fixed continuation adjustment, but the paper does not claim that an endogenous Bellman value has a Lipschitz modulus independent of η . The section assumes directly that B_A is continuous, or Lipschitz when a quantitative projection threshold is required. The one-period objective is

$$J_A^\eta(y) = B_A(y) + \eta T(x, \xi(y)), \quad y \in Y(A), \quad (5.6)$$

where $x = \Psi(A, A^*)$.

Remark 5.2 (η -dependent residual terms). *If an alternative reduced form is written as B_A^η , the finite-threshold argument requires its own Lipschitz modulus L_A^η and the inequality $\eta \underline{q}_A m_A > L_A^\eta$. A uniform large- η conclusion follows, for example, if $\limsup_{\eta \rightarrow \infty} L_A^\eta / \eta < \underline{q}_A m_A$. The paper does not assert that a continuation value generated by a full recursive addition model satisfies this condition.*

The section uses the positive-density subclass of Corollary 3.3. Given the current deviation x , let

$$I_A(x) = \text{co}(\{x\} \cup \xi(Y(A)))$$

be the compact interval spanned by the current deviation and all feasible terminal deviations, and set

$$\underline{q}_A = \min \left\{ \text{ess inf}_{z \in I_A(x)} q^+(z), \text{ess inf}_{z \in I_A(x)} q^-(z) \right\} > 0. \quad (5.7)$$

This lower bound connects the Stieltjes representation to the one-period dominance threshold: each unit of toward-target motion on the relevant path has positive value, and each unit of away-from-target motion has positive cost.

Lemma 5.1 (Distance gap of target projection). *Let $Y(A)$ be a nonempty compact interval, and let $\xi(y) = \Psi(y, a)$ be continuous and strictly monotone on $Y(A)$. Suppose there is $m_A > 0$ such that*

$$|\xi(y) - \xi(z)| \geq m_A |y - z|, \quad y, z \in Y(A). \quad (5.8)$$

Let

$$\Pi_A = \arg \min_{y \in Y(A)} |\xi(y)|.$$

Then for every $y \in Y(A)$,

$$|\xi(y)| - \min_{z \in Y(A)} |\xi(z)| \geq m_A \text{dist}(y, \Pi_A). \quad (5.9)$$

If the projection is unique, with unique projection point y^Π , then

$$|\xi(y)| - |\xi(y^\Pi)| \geq m_A |y - y^\Pi|.$$

Proof. See the projection-gap appendix. □

Lemma 5.2 (Directed recovery gap). *Under the conditions of Lemma 5.1, suppose T is given by the directed density representation in Corollary 3.3 and satisfies the positive marginal-intensity condition (5.7). Then for any $y \in Y(A)$ and any $y^\Pi \in \Pi_A$,*

$$T(x, \xi(y^\Pi)) - T(x, \xi(y)) \geq \underline{q}_A \{ |\xi(y)| - |\xi(y^\Pi)| \}. \quad (5.10)$$

If the projection is unique, then

$$T(x, \xi(y^\Pi)) - T(x, \xi(y)) \geq \underline{q}_A m_A |y - y^\Pi|. \quad (5.11)$$

Proof. See the recovery-gap appendix. □

Lemma 5.3 (Target projection maximizes directed recovery). *Suppose the conditions of Lemmas 5.1 and 5.2 hold. Then, for fixed current deviation x , $T(x, \xi(y))$ is maximized on the target-projection set Π_A . Moreover, for every open set $G \supset \Pi_A$, there exists $\Delta_G > 0$ such that*

$$T(x, \xi(y^\Pi)) - T(x, \xi(y)) \geq \Delta_G, \quad y \in Y(A) \setminus G,$$

where $y^\Pi \in \Pi_A$.

Proof. Lemma 5.2 gives a weak recovery advantage of projection points over all nonprojection points. If $G \supset \Pi_A$ is open, the compact set $Y(A) \setminus G$ is separated from Π_A . Lemma 5.1 gives some $\varepsilon_G > 0$ such that

$$|\xi(y)| - |\xi(y^\Pi)| \geq \varepsilon_G, \quad y \in Y(A) \setminus G.$$

Take $\Delta_G = \underline{q}_A \varepsilon_G$. □

5.2. Target projection under one-period dominance

Proposition 5.1 (Target projection under one-period dominance). *This proposition works in the positive-density local subclass of $\mathfrak{D}^{\text{ac}}$. Let $Y(A)$ be a nonempty compact interval, and suppose B_A and $T(x, \xi(\cdot))$ are continuous on $Y(A)$. Suppose ξ is continuous and strictly monotone on $Y(A)$, and that T satisfies the directed density representation and positive marginal-intensity condition in Lemma 5.3. Let*

$$y_\eta \in \arg \max_{y \in Y(A)} J_A^\eta(y).$$

As $\eta \rightarrow \infty$, every cluster point of any sequence $\{y_\eta\}$ belongs to the target-projection set Π_A .

Proof. See the limit-projection appendix. □

5.3. Finite-threshold target capture

Corollary 5.1 (Finite-threshold target capture). *This corollary works in the positive-density local subclass of $\mathfrak{D}^{\text{ac}}$ and assumes that the smooth term is Lipschitz on the feasible set. Consider the linear BM law $A^+ = (1 - \delta)A + c$ with $c \in [0, \bar{c}]$. Let $Y(A)$ be the compact interval above, suppose ξ is strictly increasing on $Y(A)$, and suppose there exists $m_A > 0$ such that*

$$|\xi(y) - \xi(z)| \geq m_A |y - z|, \quad y, z \in Y(A).$$

Suppose the target projection $y^\Pi(A)$ is unique and that the directional kernels satisfy the positive lower bound (5.7). If B_A is Lipschitz on $Y(A)$ with constant L_A and

$$\boxed{\eta q_{-A} m_A > L_A}, \tag{5.12}$$

then $y^\Pi(A)$ is the unique maximizer of J_A^η . Hence

$$\boxed{A^+ = y^\Pi(A)}. \tag{5.13}$$

In particular, if $A^{*+} \in Y(A)$, then

$$\boxed{A^+ = A^{*+}}. \tag{5.14}$$

Proof. See the finite-capture appendix. □

Corollary 5.2 (Uniform projection threshold on a state region). *Let $\mathcal{A}$ be a set of current states on which the conditions of Corollary 5.1 hold and the target projection is unique. Define*

$$\underline{\kappa}_{\mathcal{A}} := \inf_{A \in \mathcal{A}} q_{-A} m_A, \quad \bar{L}_{\mathcal{A}} := \sup_{A \in \mathcal{A}} L_A. \tag{5.15}$$

If $\underline{\kappa}_{\mathcal{A}} > 0$ and $\bar{L}_{\mathcal{A}} < \infty$, then every

$$\eta > \bar{\eta}_P(\mathcal{A}) := \frac{\bar{L}_{\mathcal{A}}}{\underline{\kappa}_{\mathcal{A}}} \tag{5.16}$$

makes the target projection the unique optimum for every $A \in \mathcal{A}$. On a compact region, continuity of the relevant bounds together with pointwise positivity of q_{-A} and m_A is a sufficient primitive route to these uniform inequalities.

Proof. For every $A \in \mathcal{A}$, $\eta q_{-A} m_A \geq \eta \underline{\kappa}_{\mathcal{A}} > \bar{L}_{\mathcal{A}} \geq L_A$. Corollary 5.1 therefore applies uniformly. □

Proposition 5.1 and Corollary 5.1 give the one-step choice meaning of directed recovery. If the state is already at the target and remains there, then $T(0, 0) = 0$. Starting away from the target, the feasible next-period state whose target-relative coordinate is closest to zero maximizes the directed recovery value of the current transition. When that recovery gain dominates variation in the reduced-form residual, the one-step maximizing response is characterized by the target-projection operator. If the target is reachable, that operator generates a projection plateau; if the target is not reachable, the response attaches to the feasible boundary closest to the target. The directed transition itself therefore supplies the one-step force of target projection; no flow reward at the target point is required.

5.4. Projection plateaus and reachability regions

Under the linear BM law (5.1) and $C = [0, \bar{c}]$, target projection has an explicit form. Fix the next-period target $a = A^{*+}$. The feasible set is

$$Y(A) = [(1 - \delta)A, (1 - \delta)A + \bar{c}].$$

Define the target-capture region

$$C(a) = \{A : a \in Y(A)\} = \left[\max \left\{ 0, \frac{a - \bar{c}}{1 - \delta} \right\}, \frac{a}{1 - \delta} \right]. \quad (5.17)$$

On this interval, the next-period target belongs to the current feasible set.

Proposition 5.2 (Projection plateau on the target-reachability region). *This proposition uses the positive-density dominance conditions of Corollary 5.1. Fix the target $a = A^{*+}$. Suppose the pointwise finite-threshold condition holds for every $A \in C(a)$; in particular, it is sufficient that the uniform bounds in Corollary 5.2 hold on $C(a)$ and that $\eta > \bar{\eta}_P(C(a))$. Then the unique one-step maximizing response satisfies*

$$g_\eta(A, a) = a, \quad A \in C(a). \quad (5.18)$$

The corresponding one-step consumption response is

$$c_\eta(A, a) = a - (1 - \delta)A, \quad A \in C(a), \quad (5.19)$$

and $c_\eta(A, a) \in [0, \bar{c}]$. Hence the one-step response map forms a horizontal projection plateau on $C(a)$.

Proof. See Appendix C.5. □

Outside the capture region, the target-projection response attaches to the feasible boundary closest to the target. In the strong-recovery region, the next-period state is

$$A^+ = \Pi_{[(1-\delta)A, (1-\delta)A + \bar{c}]}(a) = \min \{ \max \{ a, (1 - \delta)A \}, (1 - \delta)A + \bar{c} \}. \quad (5.20)$$

Target reachability determines exact capture; the directed recovery weight determines whether the one-step response follows the projection rule.

5.5. Conditional capture under repeated projection

Iterating the projection map gives a conditional finite-time geometric result. Consider a fixed target $a > 0$, linear dynamics, and $C = [0, \bar{c}]$. Assume

$$\bar{c} \geq \delta a. \quad (5.21)$$

This condition says that the target state itself can be maintained.

Proposition 5.3 (One-sided finite-time capture under repeated projection). *This proposition is stated for regions in which the target-projection rule is valid along the path; that rule is supported by the dominance conditions of Corollary 5.1. Suppose the target is fixed at $a > 0$, $0 < \delta < 1$, $C = [0, \bar{c}]$, and (5.21) holds. If the target-projection rule is valid along the path, then from any $A_0 > a$ there exists a finite N such that*

$$A_N \in C(a), \quad A_{N+1} = a.$$

Thus the state enters the capture region in finite time and hits the target exactly in the following period.

Proof. See the finite-time-capture appendix. □

The proposition is conditional on repeating the one-step projection map; it does not derive that map from a Bellman problem. Inside the capture region, the one-step response function directly selects the target. Outside the capture region, the projection rule pushes the state toward the target along the feasible boundary. Under that conditional iteration, target pull is expressed through the reachability region and the projection plateau.

Remark 5.3 (Zero target and the left-side modules). *In the nonnegative BM state space with $c \geq 0$, a zero target makes the physical left side economically unreachable. Moreover, with $c = 0$ and $A_0 > 0$, $A_t = (1 - \delta)^t A_0 > 0$ at every finite date. Exact finite-time capture in Proposition 5.3 therefore concerns sustainable positive treatment targets. A zero abstinence target is approached asymptotically unless the feasible technology permits an exact jump to zero.*

6. Discussion and Conclusion

The paper’s primary object is an additive measurement problem on a path-generated domain that is not closed under addition. A single continuous target-relative experience generates a four-module ledger recording direction, side, location, and multiplicity. Rational addition supplies the leading economic application, but the mathematical problem and the representation theorem are not derived from the Becker–Murphy model.

The central theoretical contribution is a boundary result for this partial domain. LC-A is a natural candidate because it permits every common formal increment whose two translations remain accessible. Nevertheless, it does not imply finite cancellation. The counterexample exploits the feature that makes the path domain nonstandard: connectivity and endpoint-flow restrictions can protect a secondary tie breaker from every admissible common shift, even though a balanced finite list reveals global inconsistency. This is why ordinary local translation reasoning cannot replace finite-list consistency.

The positive theorem is a separate sufficiency construction. Same-ledger indifference, a weak order, finite cancellation, difference completeness, list Archimedean regularity, strict directional monotonicity, and module-normalized path continuity yield an exact four-module Stieltjes representation. Finite cancellation constructs a canonical partially ordered abelian group from actual comparisons and formal balance identities. Difference completeness and list Archimedean regularity provide a transparent sufficient route to a real-valued scale, and PGM^{C+} converts positive finitely additive interval contents into Stieltjes measures. These assumptions are not claimed to be necessary, weakest, or a minimal behavioral characterization. The fixed-initial-state version of the additive-boundary problem remains open.

The representation permits direction- and side-specific measures, target-arrival atoms, and monotone coordinate reparameterizations. Target atoms are legitimate in the static representation but require an additional no-positive-loop discipline in recursive applications. The representation is covariant under monotone reparameterization: measures and densities transform, while total path values remain unchanged. The finite-interval decomposition lemma and the canonical quotient construction make explicit the bridge from actual paths to the Stieltjes integrals.

The economic sections illustrate rather than generate the theory. Conditional on a predetermined, nonbinding target schedule, the recovery wedge shows how target-relative process value changes the full consumption-stock cross effect. The one-step projection problem shows how sufficiently strong target pull can select the feasible point closest to the target and produce a projection plateau. Conditional iteration of that map gives a one-sided finite-time capture result for sustainable positive targets. None of these statements is presented as a Bellman-optimal policy theorem, and a zero abstinence target is generally approached only asymptotically under the nonnegative Becker–Murphy state law.

Several questions remain. The most important is whether economically observable comparisons restricted to a common complete initial state, together with suitable bridge conditions, can imply finite cancellation or another real-representation condition. A second question is whether difference completeness and list Archimedean regularity can be replaced by weaker, more behaviorally transparent assumptions. A third is the formation and revision of the target itself. A fourth is empirical identification of the target process, recovery coordinate, and directional measures.

The main conclusion is therefore twofold. On a genuine path-generated domain, local common-increment cancellation and global additive consistency are distinct. Once finite-list consistency and the stated sufficient regularity conditions are imposed, the resulting target-relative path order admits an exact Stieltjes representation. Rational addition gives this abstract boundary a concrete economic interpretation through directed recovery.

A. Terminology and notation

This appendix synchronizes the terminology used in the representation theorem and in the two economic applications. The primitive representation object is a weak order over single continuous target-relative paths. The induced order is defined only on ledgers accessible from such paths. Formal sums and differences state finite balance conditions and construct an ordered group; they are not additional primitive experiences.

A.1. Domain hierarchy and endpoint convention

The paper uses

$$\mathfrak{D}^{\text{smooth}} \subset \mathfrak{D}^{\text{ac}} \subset \mathfrak{D}^{\text{reg}}.$$

The regular class permits four finite Stieltjes measures with atoms. The absolutely continuous subclass admits physical toward and away densities. The smooth subclass supplies the differentiability required for the recovery-wedge

calculations. Canonical module intervals are right-closed and half-open, $(a, b]$. Incoming toward motion includes the target; outgoing away motion excludes it. Every arrival is counted once, stationary pieces have zero ledger, and no first-arrival bonus is introduced.

A.2. Core terms

Continuous path	A continuous map $\gamma : [0, T] \rightarrow X$ that is monotone on a finite partition.
Four modules	$\mathcal{M} = \{T_L, T_R, A_L, A_R\}$: toward and away motion on the two sides of the target.
Directional ledger	$\ell(\gamma) = (\tilde{h}_\gamma^{T_L}, \tilde{h}_\gamma^{T_R}, \tilde{h}_\gamma^{A_L}, \tilde{h}_\gamma^{A_R})$.
Accessible domain	$\mathcal{D} = \ell(\Gamma_F(X))$, generally not closed under addition.
Finite cancellation	No balanced finite list of actual comparisons has all left sides weakly preferred and one strictly preferred.
LC-P / LC-A	Location-free common-increment cancellation with an accessible / formal nonnegative common increment.
DC / LA	Difference completeness and list Archimedean regularity, used as sufficient real-valued regularity conditions.
PGMC⁺	Module-normalized path-generated monotone continuity.
Directed measures	The four nonnegative Stieltjes measures $\lambda_{T_L}, \lambda_{T_R}, \lambda_{A_L}, \lambda_{A_R}$.
Recovery wedge	$\mathcal{R}_d$ in $W_{cA} = u_{cA} + \eta \mathcal{R}_d$.
Target projection	A feasible next state minimizing target-relative distance.
Capture region	The current-state set from which the stated projection rule selects the target.

A.3. Main notation

Symbol	Meaning
A, A^*, c, F	Actual state, target state, action, and state transition.
$x = \Psi(A, A^*)$	Target-relative coordinate.
$\Gamma_F(X)$	Continuous finite-piecewise-monotone path domain.
$\mathcal{M}, \Xi_m$	Directional modules and their increasing coordinate intervals.
$\tau_m(a, b)$	Canonical accessible interval ledger in module m .
$\ell(\gamma), \mathcal{D}$	Path ledger and accessible ledger domain.
E, F	Ambient step-function group and accessible difference group.
P, H, G	Weak-comparison cone, indifference subgroup, and canonical ordered quotient group.
$u_0, \hat{U}, U$	Group-valued ledger, real ledger, and path representations.
v_m, G_m, λ_m	Module interval content, cumulative potential, and Stieltjes measure.
μ^T, μ^A, q^T, q^A	Side-compatible physical measures and densities.
$T(x, x')$	One-period target-relative transition payoff.
$W_{cA}, \mathcal{R}_d$	Full local cross effect and recovery wedge.
$Y(A), y^{\text{II}}(A)$	Feasible next-state set and target projection.
$C(a)$	Target-capture region.

B. Proofs for the directed recovery representation

This appendix proves the representation results in Section 3 and the boundary result in Proposition 3.3. The proofs use formal ledger sums only inside the ambient group E ; no formal sum is treated as an additional primitive path.

B.1. Reachability of finite-step ledgers

Proof of Proposition 3.1. Suppose first that $L = \ell(\gamma)$. Refine a monotone partition of γ by inserting every time at which the path meets a vertex of $\mathcal{P}$. Each resulting nonstationary subsegment traverses exactly one adjacent physical band and therefore gives one directed arc of $G(L, \mathcal{P})$. The time order of these arcs is a directed trail using every arc with its ledger multiplicity. Its used vertices are connected. Every intermediate visit contributes one incoming and one outgoing arc, while the first and last vertices contribute the imbalance in (3.8). Arcs ending at 0 are precisely the incoming toward segments, which proves the target-arrival statement.

Conversely, suppose the connectivity and degree conditions hold. The directed Euler-trail theorem gives a trail from s to t that uses every directed arc of $G(L, \mathcal{P})$ exactly once (an Euler circuit if $s = t$). Realize each arc by the

linear monotone path across its adjacent physical band and concatenate the arcs in trail order. Consecutive arcs share a vertex, so the concatenation is continuous and finite piecewise monotone. By construction, its four module traversal counts, including target arrivals under the right-closed convention, equal L . Hence $L \in \mathcal{D}$. Common refinement merely subdivides arcs and does not change the existence of the trail. $\square$

B.2. Finite cancellation and the canonical ordered group

Recall that

$$F = \langle \mathcal{D} - \mathcal{D} \rangle_{\mathbb{Z}} \subseteq E.$$

Define

$$P = \left\{ \sum_{j=1}^r (L_j - M_j) : r < \infty, \quad L_j, M_j \in \mathcal{D}, \quad L_j \geq M_j \right\} \subseteq F, \quad (\text{B.1})$$

where the empty sum is allowed, and set

$$H = P \cap (-P). \quad (\text{B.2})$$

Lemma B.1. *The set P contains zero and is closed under addition. The set H is a subgroup of F . If $\pi : F \rightarrow G := F/H$ is the quotient map and $G_+ := \pi(P)$, then*

$$G_+ + G_+ \subseteq G_+, \quad G_+ \cap (-G_+) = \{0\}.$$

Consequently,

$$g \geq h \iff g - h \in G_+. \quad (\text{B.3})$$

defines a translation-invariant partial order on G .

Proof. The empty list gives $0 \in P$, and concatenating two finite lists of weak comparisons proves $P + P \subseteq P$. If $h_1, h_2 \in H$, then $h_1, -h_2, -h_1, h_2 \in P$, so

$$h_1 - h_2 \in P \quad \text{and} \quad -(h_1 - h_2) \in P.$$

Thus H is a subgroup.

Only pointedness of the quotient cone requires an argument. Let $g \in G_+ \cap (-G_+)$. Choose $p, q \in P$ with $g = \pi(p)$ and $-g = \pi(q)$. Then $p + q \in H$, hence $-(p + q) \in P$. Since $q \in P$,

$$-p = q - (p + q) \in P.$$

Thus $p \in H$ and $g = 0$. The standard cone construction now gives reflexivity, transitivity, antisymmetry, and translation invariance of (B.3). $\square$

Proof of Proposition 3.4. Define

$$u_0 : \mathcal{D} \rightarrow G, \quad u_0(L) = \pi(L).$$

If $L \geq M$, then $L - M \in P$, and hence $u_0(L) \geq u_0(M)$.

Conversely, suppose $u_0(L) \geq u_0(M)$. There is $p \in P$ such that $\pi(L - M) = \pi(p)$. Therefore

$$h := (L - M) - p \in H \subseteq P,$$

so $L - M = p + h \in P$. Thus there are actual weak comparisons $A_j \geq B_j$ satisfying

$$L - M = \sum_j (A_j - B_j). \quad (\text{B.4})$$

If $L \not\geq M$, completeness gives $M > L$. Adding this strict comparison to the weak comparisons in (B.4) produces a balanced finite list with a strict member, contrary to finite cancellation. Hence $L \geq M$.

If $L > M$ but $u_0(L) = u_0(M)$, then $M - L \in P$. A finite weak-comparison decomposition of $M - L$, together with $L > M$, again violates finite cancellation. Strict comparisons therefore map to strict group inequalities.

Finally, every balance identity in E is preserved because π is a group homomorphism. The construction is canonical in the following sense. Suppose Q is an ordered abelian group and $v : D \rightarrow Q$ extends to a group homomorphism $\tilde{v} : F \rightarrow Q$ that sends every actual weak-comparison difference to Q_+ . If $h \in H$, then both h and $-h$ are finite sums of weak-comparison differences, so

$$\tilde{v}(h) \geq 0 \quad \text{and} \quad -\tilde{v}(h) \geq 0.$$

Antisymmetry gives $\tilde{v}(h) = 0$. Hence $H \subseteq \ker \tilde{v}$, and there is a unique positive homomorphism $\bar{v} : G \rightarrow Q$ such that $v = \bar{v} \circ u_0$. $\square$

B.3. Difference regularity and real-valued representation

For finite lists $\mathbf{A}, \mathbf{B}$ of accessible ledgers, write

$$\Delta(\mathbf{A}, \mathbf{B}) = \sum_{A \in \mathbf{A}} A - \sum_{B \in \mathbf{B}} B.$$

By the definition in Section 3.5,

$$\mathbf{A} \geq_{\Delta} \mathbf{B} \iff \Delta(\mathbf{A}, \mathbf{B}) \in P. \tag{B.5}$$

Its asymmetric part satisfies

$$\mathbf{A} >_{\Delta} \mathbf{B} \iff \pi(\Delta(\mathbf{A}, \mathbf{B})) > 0. \tag{B.6}$$

Lemma B.2. *Under difference completeness, the order on G is linear. Under list Archimedean regularity, this linear order is Archimedean: for all $g, h > 0$ in G , there exists $n \geq 1$ such that $ng \geq h$.*

Proof. Every $x \in F$ is the difference of two finite accessible-ledger lists. Difference completeness and (B.5) give $x \in P$ or $-x \in P$. Hence every $\pi(x) \in G$ is comparable with zero, and translation invariance makes the group order linear.

Let $g, h > 0$. Choose $x, y \in F$ with $\pi(x) = g$ and $\pi(y) = h$. Because $g, h > 0$, one may choose finite lists $(\mathbf{A}, \mathbf{B})$ and $(\mathbf{C}, \mathbf{D})$ such that

$$x = \Delta(\mathbf{A}, \mathbf{B}), \quad y = \Delta(\mathbf{C}, \mathbf{D}),$$

and

$$\mathbf{A} >_{\Delta} \mathbf{B}, \quad \mathbf{C} >_{\Delta} \mathbf{D}.$$

List Archimedean regularity gives an n such that

$$(n\mathbf{A}) \sqcup \mathbf{D} \geq_{\Delta} (n\mathbf{B}) \sqcup \mathbf{C}.$$

Using (B.5),

$$nx - y \in P,$$

and therefore $ng \geq h$. $\square$

Proof of Proposition 3.5. By Proposition 3.4 and Lemma B.2, G is an Archimedean linearly ordered abelian group. Hölder's theorem for ordered groups gives an order embedding

$$\varphi : G \hookrightarrow (\mathbb{R}, +)$$

unless $G = \{0\}$, in which case the zero map is sufficient. Define

$$\hat{U} = \varphi \circ u_0, \quad \tilde{U} = \varphi \circ \pi.$$

Because φ is an order embedding,

$$L \geq M \iff \hat{U}(L) \geq \hat{U}(M),$$

and $\tilde{U} : F \rightarrow \mathbb{R}$ is a group homomorphism extending $\hat{U}$.

For uniqueness, let $V_1, V_2 : D \rightarrow \mathbb{R}$ be exact representations with group-homomorphic extensions $\tilde{V}_1, \tilde{V}_2 : F \rightarrow \mathbb{R}$. Each extension maps every generator $A - B$ of P to a nonnegative number, so it maps P into $\mathbb{R}_+$. If $h \in H = P \cap (-P)$, then both h and $-h$ lie in P ; therefore $\tilde{V}_i(h) = 0$. Hence $H \subseteq \ker \tilde{V}_i$, and there is a unique homomorphism $\bar{V}_i : G \rightarrow \mathbb{R}$ satisfying $\tilde{V}_i = \bar{V}_i \circ \pi$.

The induced homomorphism $\bar{V}_i$ is an order embedding. Indeed, if $g = \pi(x) > 0$, write x as a finite sum of actual weak-comparison differences. If all corresponding numerical differences were zero, exactness would make every comparison an indifference, so $-x \in P$ and $g = 0$, a contradiction. Thus $\bar{V}_i(g) > 0$.

It remains to compare the two order embeddings $\bar{V}_1, \bar{V}_2$. Fix $a > 0$ and put $c = \bar{V}_2(a)/\bar{V}_1(a) > 0$. For every $x > 0$ and positive integers m, n ,

$$nx \leq ma \iff n\bar{V}_1(x) \leq m\bar{V}_1(a) \iff n\bar{V}_2(x) \leq m\bar{V}_2(a).$$

Hence the positive real numbers $\bar{V}_1(x)/\bar{V}_1(a)$ and $\bar{V}_2(x)/\bar{V}_2(a)$ have the same rational upper cuts and are equal. Thus $\bar{V}_2 = c\bar{V}_1$ on G , and consequently $V_2 = cV_1$ on D . This proves common positive-scale uniqueness for precisely the class stated in the proposition. $\square$

The preceding argument establishes a sufficient route to the real line. It does not make difference completeness or list Archimedean regularity necessary for every possible real-valued representation.

B.4. An explicit nonempty class with target atoms

This subsection verifies Example 3.1 and records the implications of target atoms for two-measure stitching and closed excursions. Let P and $H = P \cap (-P)$ be as in equations (B.1)–(B.2), and write

$$U_\rho := \tilde{U}_{\rho_L, \rho_R} : F \longrightarrow \mathbb{R}$$

for the group-homomorphic extension of (3.16).

Lemma B.3. *For Example 3.1,*

$$H = \ker U_\rho, \quad P = \{x \in F : U_\rho(x) \geq 0\}.$$

Proof. Every actual weak-comparison difference has nonnegative U_ρ value, so

$$P \subseteq \{U_\rho \geq 0\}, \quad H \subseteq \ker U_\rho.$$

We prove the reverse inclusions.

Call a canonical toward interval *atom-free* if its right endpoint is strictly below the target endpoint 0 in module coordinates. Any two atom-free toward intervals of equal length have the same utility, so their difference belongs to H . The same is true of any two away intervals of equal length.

Next replace a toward interval ending at the target by finitely many atom-free toward intervals. Suppose the interval lies on side $S \in \{L, R\}$, has length $s > 0$, and carries target atom ρ_S . Choose

$$0 < t < \min\{s, 2\}.$$

Cut the original interval into an atom-free interval of length $s - t$ and a final target-arrival interval of length t . The latter has utility $t + \rho_S < 4$, so the module contains an atom-free toward interval of exactly that length. The two are actual canonical paths and are indifferent. Hence, modulo H , every target-arrival interval equals a finite sum of atom-free toward intervals whose total length is exactly its U_ρ value.

Third, replace away intervals by negative atom-free toward content. Subdivide any away interval so that each piece has length s with $2s < 4$. On a physical interval strictly away from the target, move toward the target over length $2s$ and return along the same interval. The resulting closed-excursion ledger C_s does not collect a target atom and satisfies

$$U_\rho(C_s) = 2s - 2(2s) = -2s.$$

Let A_s be any away canonical interval of length s . Then $U_\rho(A_s) = -2s$, so $C_s - A_s \in H$. Split the away part of the excursion into two intervals of length s and use equal-length away indifference. This gives

$$T_{2s} + A_s \in H, \tag{B.7}$$

where T_{2s} is an atom-free toward interval of length $2s$. Thus a positive away interval is congruent modulo H to negative atom-free toward content of twice its length, with the sign reversed for a negative away interval.

Now take $x \in F$ with $U_\rho(x) = 0$. Write x as a finite signed sum of canonical intervals and subdivide as needed. The preceding relations convert x , modulo H , into a finite signed sum of atom-free toward intervals. The total positive and negative lengths coincide because their difference is $U_\rho(x) = 0$. A further finite subdivision pairs equal lengths on the two sides; every paired difference belongs to H . Hence $x \in H$, proving $\ker U_\rho \subseteq H$.

Finally take $x \in F$ with $r := U_\rho(x) > 0$. Choose n large enough that some toward module contains an atom-free canonical interval of length r/n . Let p be the formal sum of n such interval ledgers. Every term is strictly preferred to zero, so $p \in P$, and $U_\rho(p) = r$. Thus $x - p \in \ker U_\rho = H \subseteq P$, and therefore $x \in P$. Together with the zero case, this proves the result. $\square$

Verification of Example 3.1. Equation (3.16) defines a complete and transitive ledger order and satisfies same-ledger indifference. Each of the four measures assigns strictly positive normalized content to every nonempty interval, so strict directional monotonicity holds. Finite cancellation follows immediately from additivity of U_ρ .

By Lemma B.3, every $x \in F$ satisfies $x \in P$ or $-x \in P$, which proves difference completeness. If two strict list differences have U_ρ values $g, h > 0$, choose an integer n with $ng \geq h$; the same lemma gives list Archimedean regularity.

For an away module, normalized interval content is twice interval length. For a toward module on side S ,

$$\nu_{T_S}((a, b]) = b - a + \rho_S \mathbf{1}_{\{b=0\}}.$$

If $b_n \downarrow a$ and $b_n > a$, the equality $b_n = 0$ can occur only finitely often; eventually $b_n < 0$, and therefore $\nu_{T_S}((a, b_n]) = b_n - a \rightarrow 0$. Hence PGMC^+ holds. By contrast, $(-1/n, 0]$ decreases to $\{0\}$ rather than to the empty set and has content converging to ρ_S ; this is exactly how PGMC^+ permits a target atom.

The four representing measures are

$$\lambda_{T_L} = \text{Leb} + \rho_L \delta_0, \quad \lambda_{T_R} = \text{Leb} + \rho_R \delta_0, \quad \lambda_{A_L} = \lambda_{A_R} = 2 \text{Leb}.$$

They have no atoms at their normalized lower endpoints. If $\rho_L = \rho_R$, the target atoms are compatible and the two-measure form exists. If $\rho_L \neq \rho_R$ and one physical toward measure μ^T existed, left and right arrival intervals of length ε would require

$$\mu^T((-\varepsilon, 0]) = \varepsilon + \rho_L, \quad \mu^T([0, \varepsilon)) = \varepsilon + \rho_R.$$

As $\varepsilon \downarrow 0$, continuity from above of a finite measure would force both quantities to converge to $\mu^T(\{0\})$, and hence $\rho_L = \rho_R$, a contradiction. The target-atom compatibility condition is therefore indispensable.

Finally, inserting a closed excursion in an interval that does not touch the target changes value by $-r < 0$, so the model is path-sensitive. $\square$

Proposition B.1 (Positive target atoms and short closed excursions). *Under the directed-arrival convention, suppose a toward measure on one side satisfies $\lambda_T(\{0\}) = \rho > 0$, while the corresponding away measure is finite and assigns zero mass to the target. If that side permits arbitrarily short paths that touch the target and immediately return along the same interval, then some closed excursion has positive value. The conclusion remains true when the return segment is discounted.*

Proof. Let I_r be a physical interval of length r ending at the target. The incoming toward segment has value at least ρ . The away return costs $\lambda_A(I_r \setminus \{0\})$. These sets decrease to the empty set as $r \downarrow 0$, so continuity from above of the finite away measure gives

$$\lambda_A(I_r \setminus \{0\}) \longrightarrow 0.$$

For sufficiently small r , the closed excursion therefore has strictly positive net value. Multiplying the return cost by a discount factor $\beta \in (0, 1]$ can only reduce that cost. $\square$

B.5. Positive interval contents and Stieltjes measures

Fix a module m with coordinate interval $\Xi_m = [\alpha_m, \beta_m]$, and let

$$\varepsilon_m = \begin{cases} 1, & m \in \{T_L, T_R\}, \\ -1, & m \in \{A_L, A_R\}. \end{cases}$$

Define, for $\alpha_m \leq a < b \leq \beta_m$,

$$v_m((a, b]) = \varepsilon_m \hat{U}(\tau_m(a, b)). \quad (\text{B.8})$$

Lemma B.4. *The interval content v_m is strictly positive on every nonempty canonical interval and finitely additive on adjacent intervals:*

$$v_m((a, c]) = v_m((a, b]) + v_m((b, c]), \quad a < b < c.$$

Proof. Strict directional monotonicity gives $\tau_m(a, b) > o$ in a toward module and $o > \tau_m(a, b)$ in an away module. Exact representation and the sign ε_m therefore give $v_m((a, b]) > 0$.

The formal ledger identity

$$\tau_m(a, c) = \tau_m(a, b) + \tau_m(b, c)$$

holds in F . Applying the homomorphism $\tilde{U}$ and multiplying by ε_m proves finite additivity. $\square$

Lemma B.5 (PGMC continuity). *If $b_n \downarrow a$ with $b_n > a$, then*

$$v_m((a, b_n]) \longrightarrow 0.$$

Proof. By positivity and finite additivity, $v_m((a, b_n])$ is nonincreasing and has a limit $c \geq 0$. Suppose $c > 0$. Choose any nonempty canonical interval K in module m and write $V = v_m(K) > 0$. Partition K into q nonempty adjacent subintervals with $q > V/c$. Finite additivity implies that at least one subinterval J satisfies

$$0 < v_m(J) \leq V/q < c.$$

For every n ,

$$v_m((a, b_n]) \geq c > v_m(J).$$

Exact representation says

$$\tau_m(a, b_n) \succ_m^+ J \quad \text{for every } n.$$

PGMC⁺ then gives $o \succeq_m^+ J$, contradicting strict directional monotonicity. Thus $c = 0$.

The proof uses only actual canonical intervals. It does not require exact solvability. If a sequence is allowed to contain terms equal to a , the conclusion is immediate from that term onward. $\square$

Lemma B.6. *There is a unique finite Borel measure λ_m on Ξ_m , normalized by $\lambda_m(\{\alpha_m\}) = 0$, such that*

$$\lambda_m((a, b]) = v_m((a, b]) \quad (\alpha_m \leq a < b \leq \beta_m). \quad (\text{B.9})$$

Proof. Define

$$G_m(\alpha_m) = 0, \quad G_m(x) = v_m((\alpha_m, x]) \quad (x > \alpha_m).$$

Finite additivity gives

$$G_m(b) - G_m(a) = v_m((a, b]) \geq 0,$$

so G_m is nondecreasing. If $x_n \downarrow x$, then

$$G_m(x_n) - G_m(x) = v_m((x, x_n]) \longrightarrow 0$$

by Lemma B.5; hence G_m is right-continuous. The Lebesgue–Stieltjes construction on the compact interval Ξ_m produces a finite Borel measure satisfying (B.9). The normalization at α_m removes the only mass not identified by intervals of the form $(a, b]$. Such intervals form a generating semiring, so the normalized measure is unique. $\square$

Lemma B.7 (Finite interval decomposition of path ledgers). *For every $\gamma \in \Gamma_F(X)$ and every module m , there is an integer $K_m \geq 0$, canonical half-open intervals $I_{m1}, \dots, I_{mK_m} \subseteq \Xi_m$, and positive integers $n_{m1}, \dots, n_{mK_m}$ such that*

$$\tilde{h}_\gamma^m = \sum_{k=1}^{K_m} n_{mk} \mathbf{1}_{I_{mk}}. \quad (\text{B.10})$$

The sum is interpreted as zero when $K_m = 0$. The decomposition counts repeated passages by multiplicity and respects the directed-arrival convention: an incoming toward segment includes the target, whereas an outgoing away segment excludes it.

Proof. Choose a finite partition on which γ is monotone. Every nonconstant segment belongs to exactly one directional-side module. In that module's increasing coordinate, the segment traverses one canonical interval $(a, b]$ and contributes its indicator; a constant segment contributes zero. Summing over the finitely many segments gives a finite sum of interval indicators. Grouping identical intervals produces the positive integer multiplicities in (B.10). When a toward segment ends at the target, the target is its right endpoint and is included; when an away segment starts at the target, the target is the lower endpoint and is excluded. Further subdivision only replaces one indicator by the sum of adjacent canonical indicators, so the ledger and the stated convention are unchanged. $\square$

Proof of Theorem 3.1. Proposition 3.5 provides $\hat{U}$ and its homomorphic extension $\tilde{U}$. Apply Lemmas B.4–B.6 separately to the four modules.

By Lemma B.7, every component $\tilde{h}_\gamma^m$ of an accessible path ledger is a finite sum, with integer multiplicities, of canonical interval indicators. Using the ledger decomposition, the homomorphism property, and (B.9),

$$\hat{U}(\mathcal{L}(\gamma)) = \sum_{m \in \{T_L, T_R\}} \int_{\Xi_m} \tilde{h}_\gamma^m d\lambda_m - \sum_{m \in \{A_L, A_R\}} \int_{\Xi_m} \tilde{h}_\gamma^m d\lambda_m.$$

Since $U(\gamma) = \hat{U}(\mathcal{L}(\gamma))$, this is (3.18).

For fixed $\hat{U}$, the interval values in (B.8) determine each normalized measure uniquely. The common positive-scale uniqueness of $\hat{U}$ proved above gives the same common scale for all four measures. $\square$

Endpoint convention and atoms. In every module, a canonical segment is recorded on $(a, b]$ in the coordinate that increases along the direction of motion. The target is therefore the right endpoint of each toward coordinate and the left endpoint of each away coordinate. An incoming toward segment counts a target atom; a subsequent outgoing away segment does not. A path that starts at the target has no incoming arrival, and a stationary spell contributes the zero ledger. Hence each actual arrival is counted once, including repeated arrivals, and no first-arrival event is added.

Lemma B.5 does not eliminate atoms. Indeed, $(a, b_n] \downarrow \emptyset$ when $b_n \downarrow a$, whereas an atom at a is the jump $G_m(a) - G_m(a-)$ and is not contained in any of these intervals.

Proof of Corollary 3.1. Push the four module measures to their physical half-intervals. The away measures have zero mass at the target by their lower-endpoint normalization. If the left and right toward target atoms agree, define a physical toward measure by using the left measure on $[\underline{x}, 0)$, the right measure on $(0, \bar{x}]$, and inserting their common atom once at zero. Define the away measure by joining the two away measures on the open half-intervals and putting zero mass at the target. The four integrals then combine into (3.20). $\square$

Proof of Corollary 3.3. If the four physical module measures are absolutely continuous, take their Radon–Nikodym derivatives with respect to Lebesgue measure and join the left and right derivatives piecewise. Singleton masses vanish, so no target-atom compatibility condition is needed. Substitution in (3.18) gives (3.21). $\square$

B.6. A continuous-domain counterexample to LC sufficiency

This subsection proves Proposition 3.3. For the construction only, push the four ledger components to physical coordinates away from the target and write

$$h_L^T, \quad h_L^A \quad \text{on} \quad X^\circ := X \setminus \{0\}.$$

Keep the two tagged toward-arrival counts at the target in the vector

$$a_0(L) = (\tilde{h}_L^{T_R}(0), \tilde{h}_L^{T_L}(0)) \in \mathbb{N}^2.$$

Let

$$p_L = 3h_L^T - h_L^A, \quad s_L = h_L^A - h_L^T.$$

For an accessible ledger, the ordinary one-dimensional crossing identity implies

$$s_L \in S_X^R := \{0\} \cup \{\pm \rho \mathbf{1}_I : I \subseteq X \text{ is a nonempty half-open interval}\}, \quad (\text{B.11})$$

where $\rho(z) = -1$ for $z < 0$ and $\rho(z) = 1$ for $z > 0$. This follows by telescoping the ordinary upcrossing-minus-downcrossing count from the initial point to the terminal point and then reversing the sign on the left of the target.

Lemma B.8 (Collision and protected pairs). *If two accessible ledgers L, M satisfy*

$$a_0(L) = a_0(M), \quad p_L = p_M,$$

then either $L = M$ or, for a unique interval I ,

$$\{s_L, s_M\} = \{\rho \mathbf{1}_I, -\rho \mathbf{1}_I\}.$$

Moreover, for any integer-valued finite step function k , if

$$\rho \mathbf{1}_I + k \in S_X^R, \quad -\rho \mathbf{1}_I + k \in S_X^R,$$

then $k = 0$.

Proof. Pointwise on X° ,

$$h^T = \frac{p+s}{2}, \quad h^A = \frac{p+3s}{2}. \quad (\text{B.12})$$

For fixed p , two admissible state values in $\{-1, 0, 1\}$ can differ only by the pair $\{-1, 1\}$; a collision between zero and a nonzero state is impossible. The interval form in (B.11) therefore forces two distinct states to be opposite on the same interval. Equal target arrival vectors and (B.12) then give the stated ledger classification.

For the protected-pair claim, multiply both shifted states by ρ . The problem reduces to

$$\mathbf{1}_I + k' \in \{0, \pm \mathbf{1}_I\}, \quad -\mathbf{1}_I + k' \in \{0, \pm \mathbf{1}_I\},$$

where $k' = \rho k$. On I the two states must be 1 and -1 . Outside I they are equal, but a positive interval state and a negative interval state cannot share a nonzero value. Hence both vanish outside I , and $k' = 0$. $\square$

Let $\mathcal{F}$ be the group of integer-valued finite step functions on X° . For $f \neq g$, read $f - g$ from the right: let r be the right endpoint of its rightmost nonzero constancy interval, and declare $f >_R g$ when the constant value immediately to the left of r is positive. This is a translation-invariant strict total order and extends pointwise strict dominance.

Fix

$$0 < a < b < c < \bar{x}, \quad I_1 = (a, b], \quad I_2 = (b, c], \quad I_{12} = (a, c].$$

For every nonempty half-open interval I , set

$$\eta(I) = \begin{cases} -1, & I = I_{12}, \\ 1, & I \neq I_{12}. \end{cases}$$

Define

$$\tau(0) = 0, \quad \tau(\eta(I)\rho\mathbf{1}_I) = 1, \quad \tau(-\eta(I)\rho\mathbf{1}_I) = 0.$$

Order accessible ledgers lexicographically by

$$\Phi(L) = (a_0(L), p_L, \tau(s_L)), \tag{B.13}$$

using the ordinary lexicographic order on $a_0(L)$, the order $>_R$ on p_L , and the ordinary order on the final bit.

Proof of Proposition 3.3. Lemma B.8 shows that the key (B.13) separates any two accessible ledgers: when the first two entries coincide, distinct ledgers form exactly one opposite-state pair, and τ separates that pair. Thus Φ defines a strict total ledger order. Pulling it back through ℓ gives a path weak order satisfying same-ledger indifference.

If L has weakly more toward content and weakly less away content than M , with a strict difference, then either the target-arrival vector improves or

$$p_L - p_M = 3(h_L^T - h_M^T) + (h_M^A - h_L^A)$$

is a nonzero nonnegative step function. Hence $\Phi(L) > \Phi(M)$, proving strict directional monotonicity.

Now let $K \in E_+$ and suppose $L, M, L+K, M+K$ are accessible. A common increment translates a_0 and p by the same amounts. Therefore any strict difference in the first two entries of Φ is preserved. If those entries tie and $L \neq M$, Lemma B.8 gives opposite states $\pm\rho\mathbf{1}_I$. Accessibility of both translated ledgers forces the common state increment to be zero, so τ is unchanged. This proves LC-A.

Finally, use two adjacent right-side bands I_1, I_2 and coordinates (T_1, T_2, A_1, A_2) . The following ledgers are accessible:

$$\begin{aligned} (1, 1, 0, 0) &> (2, 2, 3, 3), \\ (0, 2, 0, 3) &> (0, 1, 0, 0), \\ (2, 0, 3, 0) &> (1, 0, 0, 0). \end{aligned} \tag{B.14}$$

For example, representatives for the three left–right pairs are

$$c \rightarrow a \quad \text{and} \quad a \rightarrow c \rightarrow a \rightarrow c \rightarrow a \rightarrow c,$$

$$b \rightarrow c \rightarrow b \rightarrow c \rightarrow b \rightarrow c \quad \text{and} \quad c \rightarrow b,$$

and

$$a \rightarrow b \rightarrow a \rightarrow b \rightarrow a \rightarrow b \quad \text{and} \quad b \rightarrow a,$$

where each arrow is the monotone piecewise-linear segment between its endpoints. In the first comparison the primary functions coincide and the favored state is $-\mathbf{1}_{I_{12}}$; in the second and third comparisons the favored states are $\mathbf{1}_{I_2}$ and $\mathbf{1}_{I_1}$. These are exactly the orientations specified by η . The two sides of (B.14) both sum to

$$(3, 3, 3, 3).$$

Thus the three strict comparisons form a finite-cancellation violation. □

$$L_i > M_i \ (i = 1, 2, 3), \quad \sum_i L_i = \sum_i M_i = (3, 3, 3, 3)$$

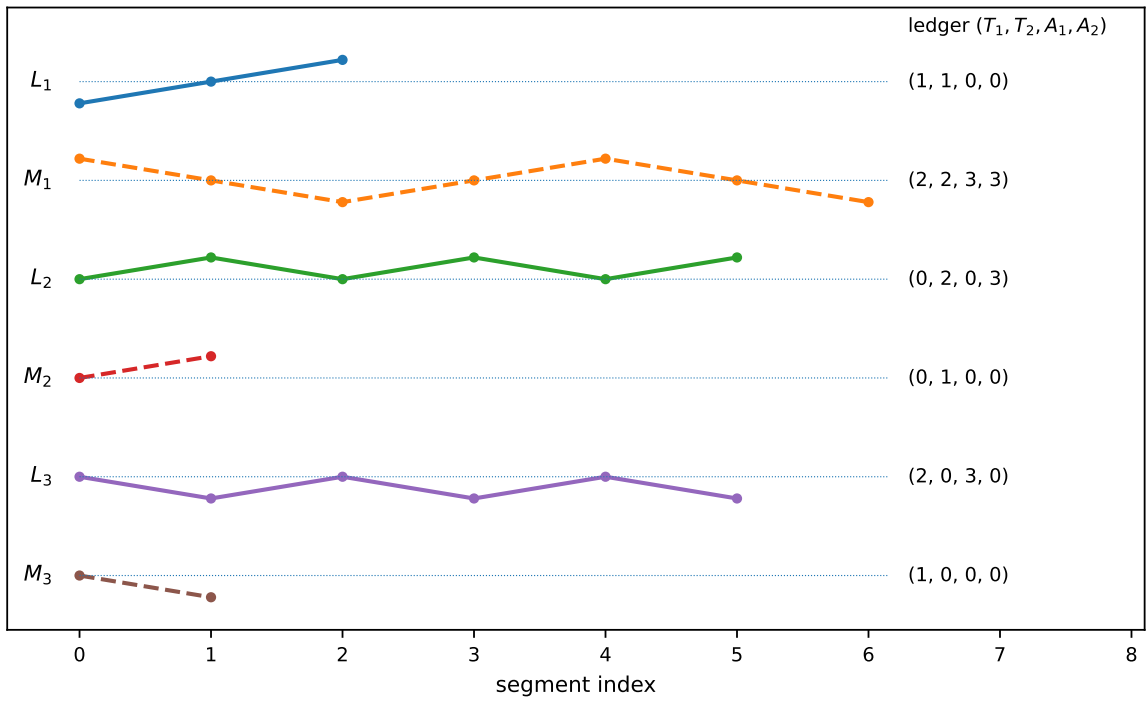


Figure 3: Representative walks for the balanced certificate C_* . Each row plots the target-side state against segment index. Solid rows are the left alternatives L_i and dashed rows are the right alternatives M_i . The three comparisons satisfy $L_i > M_i$, while both ledger sums equal $(3, 3, 3, 3)$; hence the list violates finite cancellation.

C. Proofs of the main results

This appendix proves the results in Sections 4–5. The order follows the theoretical chain in the text: projection geometry, the directed recovery gap, one-period target projection, projection plateaus, conditional repeated-projection capture, and the recovery wedge.

C.1. Distance gap for target projection

Proof of Lemma 5.1. Since $Y(A)$ is a compact interval and $y \mapsto |\xi(y)|$ is continuous, the projection set Π_A is nonempty and compact. Let

$$d_0 = \min_{z \in Y(A)} |\xi(z)|.$$

Because ξ is strictly monotone, the projection set is a singleton. For $y \in Y(A)$, let $\hat{y} \in \Pi_A$ be the closest projection point to y .

If $0 \in \xi(Y(A))$, then $d_0 = 0$ and any projection satisfies $\xi(\hat{y}) = 0$. By the lower slope condition,

$$|\xi(y)| - d_0 = |\xi(y) - \xi(\hat{y})| \geq m_A |y - \hat{y}| = m_A \text{dist}(y, \Pi_A).$$

If $\xi(Y(A))$ lies strictly on one side of zero, then $|\xi(\cdot)|$ is monotone on the whole interval and the projection is the boundary point with minimal target distance. Along the interval between y and $\hat{y}$, the sign is fixed, so

$$|\xi(y)| - |\xi(\hat{y})| = |\xi(y) - \xi(\hat{y})| \geq m_A |y - \hat{y}| = m_A \text{dist}(y, \Pi_A).$$

The cases cover all possibilities, proving the projection-gap inequality. $\square$

C.2. Directed recovery gap

Proof of Lemma 5.2. Write $z = \xi(y)$ and $z^\Pi = \xi(y^\Pi)$. By definition of the projection, $|z^\Pi| \leq |z|$. If $z = z^\Pi$, the result is immediate. Otherwise $|z| > |z^\Pi|$.

Because $\xi(Y(A))$ is an interval and z^Π minimizes absolute value on that interval, either $z^\Pi = 0$, or z and z^Π have the same sign and z^Π lies between z and zero. Compare the two paths $x \rightarrow z$ and $x \rightarrow z^\Pi$. After their common directional ledger is cancelled, the residual consists of a toward-target segment present in the path ending at z^Π , an away-from-target segment present in the path ending at z , or both. The total length of these favorable residual pieces is exactly $|z| - |z^\Pi|$. Since both active directional densities are bounded below by $\underline{q}_A$ on the relevant compact interval, replacing the terminal point z by the closer point z^Π increases net directed recovery value by at least

$$\underline{q}_A (|z| - |z^\Pi|).$$

The recovery-gap inequality follows, and combining it with Lemma 5.1 gives the distance version. $\square$

C.3. Proof of limiting target projection

Proof. Let G be any open set containing Π_A . Since $Y(A)$ is compact and B_A is continuous, there is a finite M_G such that

$$|B_A(y) - B_A(z)| \leq M_G, \quad y, z \in Y(A).$$

For any $y \in Y(A) \setminus G$ and $y^\Pi \in \Pi_A$, the recovery-gap lemma gives

$$T(x, \xi(y^\Pi)) - T(x, \xi(y)) \geq \Delta_G > 0.$$

Thus

$$J_A^\eta(y^\Pi) - J_A^\eta(y) = B_A(y^\Pi) - B_A(y) + \eta \{T(x, \xi(y^\Pi)) - T(x, \xi(y))\} \geq -M_G + \eta \Delta_G.$$

For $\eta > M_G/\Delta_G$, no point outside G can be optimal. Since G is an arbitrary open neighborhood of Π_A , every cluster point of optimal choices as $\eta \rightarrow \infty$ belongs to Π_A . $\square$

C.4. Proof of finite-threshold target capture

Proof. Assume that the target projection is unique and denote it by $y^\Pi(A)$. By the projection-gap and recovery-gap lemmas, for every $y \in Y(A)$,

$$T(x, \xi(y^\Pi)) - T(x, \xi(y)) \geq \underline{q}_A m_A |y - y^\Pi|.$$

If B_A has Lipschitz constant L_A , then

$$B_A(y) - B_A(y^\Pi) \leq L_A |y - y^\Pi|.$$

Therefore the full one-period objective $J_A^\eta(y) = B_A(y) + \eta T(x, \xi(y))$ satisfies

$$J_A^\eta(y^\Pi) - J_A^\eta(y) \geq (\eta \underline{q}_A m_A - L_A) |y - y^\Pi|.$$

If $\eta \underline{q}_A m_A > L_A$, every $y \neq y^\Pi$ is strictly dominated by y^Π , so the projection is the unique optimum. If $A^{*+} \in Y(A)$, then $\xi(A^{*+}) = 0$ and the projection is A^{*+} . $\square$

C.5. Proof of the projection plateau and capture region

Proof. Fix $a = A^{*+}$. By definition, $A \in C(a)$ if and only if

$$a \in [(1 - \delta)A, (1 - \delta)A + \bar{c}].$$

The consumption that implements a is

$$c = a - (1 - \delta)A \in [0, \bar{c}].$$

If the finite-threshold condition holds, the unique optimal next-period state is the target projection. Since $a \in Y(A)$ and $\xi(a) = 0$, the projection is a . Hence the repeated-projection map satisfies $g_\eta(A, a) = a$, and the one-step consumption response is $c_\eta(A, a) = a - (1 - \delta)A$. $\square$

C.6. Proof of one-sided finite-time capture

Proof. Consider $A_0 > a$. If $A_t \in C(a)$, Proposition 5.2 gives $A_{t+1} = a$. If $A_t \notin C(a)$ and $A_t > a$, then $a < (1 - \delta)A_t$, so the target is below the feasible interval and the projection rule chooses the left endpoint:

$$A_{t+1} = (1 - \delta)A_t.$$

Before reaching the capture region, $A_t = (1 - \delta)^t A_0$. Let N be the smallest integer such that

$$(1 - \delta)^N A_0 \leq \frac{a}{1 - \delta}.$$

If $N = 0$, the initial state already satisfies the upper capture bound. If $N \geq 1$, minimality implies $(1 - \delta)^{N-1} A_0 > a/(1 - \delta)$, hence $A_N = (1 - \delta)^N A_0 > a$. Since $\bar{c} \geq \delta a$, the lower bound of the capture region is no greater than a . Thus $A_N \in C(a)$, and the next period satisfies $A_{N+1} = a$. $\square$

C.7. Endpoint-integral proof of the recovery wedge

Proof. Work in a nonboundary, noncrossing, direction-fixed local region. Let

$$x = \Psi(A, A^*), \quad x' = p(A') = \Psi(A', A^{*'}), \quad A' = (1 - \delta)A + c.$$

The current endpoint x is not affected by current consumption c , so the cA cross term comes from the next endpoint x' . Let $s = \text{sgn}(x')$ and let $d \in \{+, -\}$ denote the active local direction. By the Leibniz rule for endpoint integrals,

$$T_{x'}(x, x') = -sq^d(x').$$

Moreover,

$$x'_c = p'(A'), \quad x'_A = (1 - \delta)p'(A'), \quad x'_{cA} = (1 - \delta)p''(A').$$

The chain rule gives

$$T_{cA} = T_{x'x'}x'_c x'_A + T_{x'A}x'_{cA} = -s(1 - \delta) \left[q^{d'}(x')(p'(A'))^2 + q^d(x')p''(A') \right].$$

Since $W(c, A) = u(c, A) + \eta T(x, x')$, Proposition 4.1 follows. At $x' = 0$, at direction switches, or at crossing boundaries, the displayed smooth derivative formula is replaced by the corresponding one-sided statement. $\square$

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

No data were used for the research described in the article.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI's ChatGPT for translation assistance, editorial revision, consistency and derivation checks, and drafting code for schematic figures. The author reviewed and edited the output as needed, verified the mathematical content, and takes full responsibility for the article.

References

- Afriat, S. N. (1967). The construction of utility functions from expenditure data. *International Economic Review*, 8(1), 67–77. doi:10.2307/2525382.
- Becker, G. S., and K. M. Murphy (1988). A theory of rational addiction. *Journal of Political Economy*, 96(4), 675–700. doi:10.1086/261558.
- Becker, G. S., M. Grossman, and K. M. Murphy (1994). An empirical analysis of cigarette addiction. *American Economic Review*, 84(3), 396–418. doi:10.2307/2118059.
- Bernheim, B. D., and A. Rangel (2004). Addiction and cue-triggered decision processes. *American Economic Review*, 94(5), 1558–1590. doi:10.1257/0002828043052222.
- Constantinides, G. M. (1990). Habit formation: A resolution of the equity premium puzzle. *Journal of Political Economy*, 98(3), 519–543. doi:10.1086/261693.
- Clarke, F. H. (1983). *Optimization and Nonsmooth Analysis*. Wiley, New York.
- Fishburn, P. C. (1970). *Utility Theory for Decision Making*. Wiley, New York.
- Giné, X., D. Karlan, and J. Zinman (2010). Put your money where your butt is: A commitment contract for smoking cessation. *American Economic Journal: Applied Economics*, 2(4), 213–235. doi:10.1257/app.2.4.213.
- Gruber, J., and B. Köszegi (2001). Is addiction “rational”? Theory and evidence. *Quarterly Journal of Economics*, 116(4), 1261–1303. doi:10.1162/003355301753265570.
- Gul, F., and W. Pesendorfer (2001). Temptation and self-control. *Econometrica*, 69(6), 1403–1435. doi:10.1111/1468-0262.00252.
- Hindy, A., C.-F. Huang, and D. M. Kreps (1992). On intertemporal preferences in continuous time: The case of certainty. *Journal of Mathematical Economics*, 21(5), 401–440. doi:10.1016/0304-4068(92)90032-3.
- Hsiaw, A. (2013). Goal-setting and self-control. *Journal of Economic Theory*, 148(2), 601–626. doi:10.1016/j.jet.2012.08.001.
- Kahneman, D., and A. Tversky (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291. doi:10.2307/1914185.
- Koch, A. K., and J. Nafziger (2011). Self-regulation through goal setting. *Scandinavian Journal of Economics*, 113(1), 212–227. doi:10.1111/j.1467-9442.2010.01641.x.
- Köszegi, B., and M. Rabin (2006). A model of reference-dependent preferences. *Quarterly Journal of Economics*, 121(4), 1133–1165. doi:10.1093/qje/121.4.1133.
- Köszegi, B., and M. Rabin (2009). Reference-dependent consumption plans. *American Economic Review*, 99(3), 909–936. doi:10.1257/aer.99.3.909.
- Krantz, D. H., R. D. Luce, P. Suppes, and A. Tversky (1971). *Foundations of Measurement, Vol. I: Additive and Polynomial Representations*. Academic Press, New York.
- Laibson, D. (1997). Golden eggs and hyperbolic discounting. *Quarterly Journal of Economics*, 112(2), 443–478. doi:10.1162/003355397555253.
- Luce, R. D., and J. W. Tukey (1964). Simultaneous conjoint measurement: A new type of fundamental measurement. *Journal of Mathematical Psychology*, 1(1), 1–27. doi:10.1016/0022-2496(64)90015-X.
- O'Donoghue, T., and M. Rabin (1999). Doing it now or later. *American Economic Review*, 89(1), 103–124. doi:10.1257/aer.89.1.103.
- Orphanides, A., and D. Zervos (1998). Myopia and addictive behaviour. *Economic Journal*, 108(446), 75–91. doi:10.1111/1468-0297.00274.
- Pollak, R. A. (1970). Habit formation and dynamic demand functions. *Journal of Political Economy*, 78(4), 745–763. doi:10.1086/259667.
- Simon, H. A. (1955). A behavioral model of rational choice. *Quarterly Journal of Economics*, 69(1), 99–118. doi:10.2307/1884852.
- Stigler, G. J., and G. S. Becker (1977). De gustibus non est disputandum. *American Economic Review*, 67(2), 76–90. doi:10.2307/1807222.
- Strotz, R. H. (1955–1956). Myopia and inconsistency in dynamic utility maximization. *Review of Economic Studies*, 23(3), 165–180. doi:10.2307/2295722.
- Suranovic, S. M., R. S. Goldfarb, and T. C. Leonard (1999). An economic theory of cigarette addiction. *Journal of Health Economics*, 18(1), 1–29. doi:10.1016/S0167-6296(98)00037-X.
- Varian, H. R. (1982). The nonparametric approach to demand analysis. *Econometrica*, 50(4), 945–973. doi:10.2307/1912771.
- Winston, G. C. (1980). Addiction and backsliding: A theory of compulsive consumption. *Journal of Economic Behavior & Organization*, 1(4), 295–324. doi:10.1016/0167-2681(80)90009-8.
- Fuchs, L. (1963). *Partially Ordered Algebraic Systems*. Pergamon Press, Oxford.
- Scott, D. (1964). Measurement structures and linear inequalities. *Journal of Mathematical Psychology*, 1(2), 233–247. doi:10.1016/0022-2496(64)90002-1.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: